

Transparency vs. Liquidity Provision in Perpetual Futures Markets: A Market Design Framework with Hidden Liquidity

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Abstract

Perpetual futures contracts dominate cryptocurrency trading, yet the widespread use of hidden order types (iceberg orders, reserve liquidity, and non-displayed depth) has outpaced formal market design. This paper develops a complete, implementable framework for designing perpetual futures markets with both visible and hidden liquidity. We extend the Kyle (1985) batch-auction model to two layers of depth and derive in closed form how equilibrium price impact, realised spreads, and post-trade reversion depend on the hidden-to-visible ratio κ . As a foundational structural result, we establish a model-independent capacity bound on κ derived from the marginal liquidity provider's participation constraint: any two-layer matching engine has a finite *structural ceiling* $\bar{\kappa}$ above which the visible book collapses, and any welfare-derived optimum must lie below this bound. From the same primitives — together with an auxiliary Pareto assumption for the trade-size distribution beyond visible depth — we construct informativeness $I(\kappa)$ and execution quality $E(\kappa)$ in closed form, and combine them under a posited regulator welfare function (a weighted sum of the two externalities, following the market-design tradition of Glosten 1994). We prove existence of an interior optimal κ^* in the normal regime. Our central new result is that κ^* is state-dependent: in normal markets we obtain a finite interior optimum ($\kappa^* \approx 1.56$ under canonical-calibration primitives at the neutral regulator weight); under liquidation cascades, when the order-flow distribution becomes fat-tailed and price discovery is mechanically distorted, the welfare function becomes monotone increasing in κ , implying the cap should be relaxed to the highest level permitted by operational constraints (per-trader limits, refill throttles, anti-gaming rules in §6). An empirical calibration on 18 hours of Bybit BTC-PERP data (Appendix C) finds that the venue's actual microstructure ($D_v / \sigma_q \approx 0.094$, well below the canonical interior-optimum band) places the welfare optimum at the corner solution even at the neutral baseline; this is consistent with the model's qualitative predictions but indicates that the canonical $\kappa^* = 1.56$ number is venue-dependent, not universal. We supply a level-by-level implementation rule, a six-step calibration procedure using only public exchange data, and two default values for exchanges that cannot perform calibration: $\kappa^* = 1.0$ (calibration-anchored, capturing 99% of optimal welfare) or $\kappa^* = 0.5$ (minimal-intervention, for venues currently operating with no hidden depth).

Keywords: perpetual futures, market microstructure, hidden liquidity, iceberg orders, price discovery, market design, liquidation cascades.

What this paper does not claim

Several aspects of the framework are easily over-read; we list them here in the interest of claims discipline.

1. The headline value $\kappa^* \approx 1.56$ is not a universal recommendation. It is the welfare-maximising hidden-to-visible ratio under the canonical calibration of Appendix A (a stylised parameterisation with $D_v = 5$, $\sigma_q =$

$\sqrt{50}$, $\xi = 2.5$, neutral regulator weight). On real venues with shallow visible books relative to flow — which Appendix C documents for Bybit BTC-PERP — the welfare optimum at the neutral baseline is at the corner solution, not at 1.56. The number 1.56 should be read as an illustrative benchmark for the canonical calibration, useful for verifying the model's interior-optimum result, not as a prescription venues should aim for.

2. The structural ceiling $\bar{\kappa}$ of Proposition 0 is an informal participation-constraint argument, not a fully-pinned-down theorem. We sketch the proof from the marginal liquidity provider's break-even condition and identify the existence and finiteness of $\bar{\kappa}$ as robust to microstructure modelling choices. We do not provide a fully formal proof in a specific equilibrium concept, and the exact value of $\bar{\kappa}$ depends on parameters we do not derive (spread, adverse-selection intensity, flow distribution). Readers should treat Proposition 0 as a structural intuition with broad applicability rather than a theorem in the strong sense.

3. The toxicity safeguard of §6.7 is an engineering heuristic, not a theorem-backed component. The reversion-based safeguard provides operational robustness against adversarial use of hidden orders, but it is not derived from the welfare model of §5. It should be read as a sensible engineering rule whose parameter ($\gamma = 0.5$) is chosen for fail-safe behaviour rather than welfare optimality. A full equilibrium analysis incorporating informed-trader strategic placement of hidden orders is left for future work.

4. The Pareto assumption for trade-size tails is doing significant work. The closed-form expression for $E(\kappa)$ in §5.2 follows from assuming that conditional on $|q| > D_v$, the trade-size distribution is Pareto with shape ξ . This is a well-documented stylised feature of crypto order flow, but it is not derived from the model's primitives. Quantitative results — particularly the location of the threshold ω_{PD}^* between interior and corner regimes — are sensitive to ξ , and venues with non-Pareto tails (e.g., bimodal trade-size distributions during specific institutional flows) may produce different optima.

5. The empirical calibration on Bybit (Appendix C) supports the qualitative story, not the precise quantitative recommendations. The 18-hour window is short for a calibration intended to inform venue policy; the exercise is best read as a proof-of-tractability and a reality check on the canonical numbers, not as a venue-policy recommendation. Cross-venue comparison and longer windows are the natural next steps. With these caveats stated explicitly, the paper's contributions are: (i) a tractable framework relating hidden depth to price discovery and execution quality, (ii) a state-dependent design rule with a corner solution under cascade stress, (iii) a per-price-level structural ceiling on κ from LP participation, (iv) implementation rules with anti-gaming controls, and (v) a calibration procedure illustrated on real exchange data.

1. Introduction

Perpetual futures ("perps") have become the dominant derivative instrument in cryptocurrency markets, with daily volumes routinely exceeding spot markets on major exchanges. Unlike traditional futures, perps have no expiry; a periodic funding rate anchors the contract price to an underlying spot index. Recent innovations on decentralised exchanges such as dYdX v4, GMX, and Hyperliquid have introduced market designs that depart from the fully visible central limit order book. The most consequential departure is the use of hidden liquidity, which encompasses reserve orders, iceberg orders, and non-displayed depth invisible to other traders.

This raises a fundamental question: how does hidden liquidity affect price discovery, execution costs, and optimal market design in perpetual futures markets?

The answer is not obvious. Hidden depth absorbs large orders without immediate price disclosure, lowering execution costs for liquidity demanders; but the same mechanism makes observable order flow a worse signal of fundamentals, degrading price discovery. The exchange must trade these forces off, and the optimal balance turns out to depend not only on preference parameters but on market conditions. The most relevant of these is the prevalence of forced-liquidation flow, which is the defining feature distinguishing perpetual futures from spot markets.

This paper makes four contributions:

1. A two-layer Kyle model with closed-form structural welfare. We extend the Kyle (1985) batch-auction model to visible depth D_v and hidden depth D_h . We derive the equilibrium price impact $\lambda(D_h)$ in closed form. We then construct (rather than postulate the shape of) both the informativeness function $I(\kappa)$ and the execution-quality function $E(\kappa)$ from the model primitives $\beta, \sigma_v^2, \sigma_{ib}^2, D_v, D_h$, together with one auxiliary structural assumption: that the conditional distribution of trade sizes given exceedance of visible depth is Pareto with regime-dependent shape ξ . This Pareto assumption is well-documented empirically in crypto order flow and yields a tractable closed form for E ; we treat ξ as a calibrable parameter, not a model output. The welfare function combining I and E is then posited as a weighted sum (a normative choice in the market-design tradition, not a derivation), with policy weight ω_{PD} left as a regulator's parameter.

2. State-dependent optimal design. Under a regime variable s capturing liquidation-cascade stress, we prove $\kappa^*(s)$ is monotone increasing in stress (Proposition 6). Hidden depth is more valuable precisely when it is most demanded: during cascades, the order-flow distribution is fat-tailed and price discovery is mechanically distorted, so execution-quality concerns dominate informativeness concerns. This justifies a circuit-breaker-style relaxation of disclosure requirements during stress.

3. A level-by-level implementation rule. We provide explicit order types, priority rules, and the constraint $D_h(p, s) \leq \kappa^*(s) \cdot D_v(p)$. The design integrates with perpetual futures features (funding and liquidation) without modifying the core tradeoff.

4. A calibration procedure on public data. A six-step procedure recovers κ^* from top-of-book snapshots and trade tape alone. For exchanges that cannot calibrate, we recommend a calibration-anchored default $\kappa^* = 1.0$ in the normal regime, capturing 99% of the welfare available at the calibrated optimum, with a more conservative $\kappa^* = 0.5$ starting point for venues currently operating with no hidden depth.

The key insight is that visible order book depth is no longer a sufficient statistic for liquidity. A shallow visible book may conceal substantial hidden depth, creating a "liquidity illusion" that distorts trading strategies and complicates regulation. The illusion is most acute, and most consequential, during periods of leveraged-position liquidation.

1.1 Related literature

Asymmetric information and price impact. The foundational microstructure framework is Kyle (1985), which our model directly extends, and Glosten and Milgrom (1985), which derives bid-ask spreads from sequential trading with informed agents. Easley and O'Hara (1992) formalise how trading intensity reveals information; their distinction between informed and uninformed flow underlies our derivation of $I(\kappa)$.

Order book transparency. *Foucault, Pagano, and Röell (2013)* survey the theoretical and empirical literature on pre-trade transparency, including hidden orders. *Boulatov and George (2013)* study a market in which liquidity providers choose between displayed and hidden orders, and show that hiding can improve market quality by encouraging informed traders to provide liquidity rather than demand it. *Buti, Rindi, and Werner (2017)* model dark-pool trading strategies and document how hidden venues interact with limit order book quality. We complement this work by providing the implementable design rule for on-exchange hidden depth that these models lack, and by extending the analysis to perpetual derivatives.

Empirical microstructure of execution. *Hendershott and Riordan (2013)* and *Brogaard, Hendershott, and Riordan (2014)* document how algorithmic and high-frequency traders interact with visible depth and contribute to price discovery, with implications for hidden-order detection. These results motivate our calibration procedure, which uses observable execution shortfalls to back out implied $\hat{\kappa}$.

Crypto and DeFi market design. *Capponi and Jia (2025)* analyse liquidity provision on automated-market-maker exchanges, deriving conditions under which arbitrage rents accrue to validators rather than liquidity providers. *Lehar and Parlour (2025)* empirically and theoretically compare Uniswap to a centralised limit order book and identify conditions under which the AMM design dominates. Our paper differs in that it analyses a central limit order book with hidden depth, which is the dominant architecture on perpetual derivatives venues such as dYdX v4 and Hyperliquid, rather than constant-function market makers.

To our knowledge, this is the first paper to combine a structural microstructure model of hidden liquidity with an implementable, calibrated design rule for perpetual futures, and the first to formalise state-dependence of the optimal hidden ratio under liquidation stress.

1.2 Existing perpetual futures venues

To ground the design problem in current practice, we survey the architecture of six representative perpetual futures venues. Three are decentralised (dYdX v4, Hyperliquid, GMX), two are centralised (Binance Futures, Bybit Perp), and one (Aster) is a smaller DeFi venue that supports hidden orders explicitly. The dimensions of comparison are the matching architecture, whether the venue supports hidden or iceberg orders, the disclosure rule applied (if any), and how cascade conditions are handled. Table 1 summarises the findings.

Venue	Architecture	Hidden orders	Iceberg orders	Disclosure rule	Cascade handling
dYdX v4	On-chain CLOB (Cosmos appchain)	No	No	Full transparency; book is public via Indexer API	Liquidation engine; no formal circuit breaker
Hyperliquid	On-chain CLOB (custom L1, HyperBFT)	No (explicitly disclaimed)	No	Full transparency; entire book on consensus state	Partial liquidation, ADL, mark-price triggers
GMX	Oracle-priced AMM (no order book)	Not applicable	Not applicable	Pricing via Chainlink oracle; no resting orders	Liquidation against GLP/GM pools
Aster	CLOB perp DEX	Yes (full anonymity)	Yes	Documented as a trader-facing feature; no published cap	Liquidation engine; no published cascade rule

Venue	Architecture	Hidden orders	Iceberg orders	Disclosure rule	Cascade handling
Binance Futures	Centralised CLOB	No (perp; spot supports)	No (perp; spot supports max 10 splits)	Centralised order book with public top-of-book	Insurance fund, ADL, mark-price triggers
Bybit Perp	Centralised CLOB	No	Yes (max 100x order size, sub-orders)	Iceberg sub-order placement; total size hidden	Insurance fund, ADL, mark-price triggers

Table 1. Comparison of perpetual futures venues. Findings drawn from each venue's official documentation as of April 2026. "Hidden orders" refers to fully non-displayed orders; "iceberg orders" refers to orders that display only a fraction of total size at any time.

Two patterns emerge. First, the largest decentralised perpetual venues by volume (dYdX v4, Hyperliquid, GMX) have no formal support for hidden depth at all. Hyperliquid's documentation explicitly states that the matching engine maintains no hidden book. dYdX v4 supports six order types (market, limit, stop-market, stop-limit, take-profit, IoC, post-only) but none with non-displayed quantity. GMX has no order book in the conventional sense, since it prices trades against an oracle and a passive liquidity pool. The two centralised venues we examine (Binance Futures and Bybit Perp) support iceberg-style splitting on perps but not pure hidden orders, and the smaller decentralised venue Aster is the only example of explicit hidden-order support on a perp DEX.

Second, none of these venues publishes a quantitative disclosure rule. The closest analogue is Bybit's iceberg order, which caps total order size at 100 times the per-sub-order maximum, but this is a *trader-facing* parameter rather than a *venue-imposed* constraint on aggregate hidden depth at a price level. Cascade handling is similarly ad hoc: each venue uses some combination of partial liquidation, auto-deleveraging, and mark-price-based triggers, but none ties the disclosure regime to the cascade state in a formally specified way.

This is the gap our paper fills. The design rule $D_h(p, s) \leq \kappa^*(s) \cdot D_v(p)$ provides a venue-imposed, level-by-level constraint on hidden depth that is calibrated to the regime. The paper's contribution can therefore be read as *prescriptive* relative to existing practice: it tells venues without hidden orders (dYdX, Hyperliquid) what their cap *would* be if they introduced one, and tells venues with hidden orders (Aster, Bybit) what cap they *should* enforce given measured stress conditions.

1.3 Roadmap

Section 2 presents the model. Section 3 derives equilibrium with hidden liquidity, including the structural-ceiling result (Proposition 0). Section 4 develops additional microstructure results, including spread compression and post-trade reversion. Section 5 constructs the welfare components from primitives (with an auxiliary Pareto assumption for trade sizes) and solves the optimal design problem in the normal regime. Section 6 specifies the mechanism implementation: order types, priority rules, and the level-by-level disclosure constraint. Section 7 applies the framework to perpetual futures, deriving the state-dependent optimum under liquidation cascades. Section 8 lists empirical implications. Section 9 contains a diagrammatic summary. Section 10 concludes. Appendix A contains all formal propositions, proofs, and the calibration procedure. Appendix B develops a Stackelberg adversary-cost framework for deriving the anti-gaming parameters of §6.6, with the multi-account-evasion subcase worked out in full. Appendix C reports

a full empirical calibration on Bybit BTC-PERP and contextualises the canonical numerical values against actual venue microstructure.

2. Model

We consider a single-period market for a perpetual futures contract. There are three participant types: an informed trader, noise traders, and competitive market makers.

Fundamental value

The underlying spot value v (e.g., a Bitcoin index) is normally distributed:

$$v \sim N(0, \sigma_v^2).$$

Noise traders

Uninformed order flow is given by

$$u_t \sim N(0, \sigma_u^2),$$

independent of v .

Informed trader

The informed trader observes v and submits

$$x_t = \beta v, \quad \beta > 0.$$

Total order flow

$$q_t = x_t + u_t = \beta v + u_t.$$

Liquidity structure

The limit order book has two layers:

- **Visible depth** D_v : displayed limit orders, observable to all traders.
- **Hidden depth** D_h : non-displayed orders such as iceberg or reserve orders, observable only to the exchange matching engine.

The *hidden-to-visible ratio* is the central design variable of the paper:

$$\kappa \equiv D_h / D_v.$$

Equivalently, $D_h = \kappa \cdot D_v$. The exchange's design choice is κ — the cap on hidden depth as a fraction of visible depth at each price level. The optimal κ^* is the central object of §5; its state-dependent generalisation $\kappa^*(s)$ is the central object of §7. Throughout, $\kappa = 0$ corresponds to a fully transparent (visible-only) book; $\kappa \rightarrow \infty$ corresponds to all liquidity hidden.

Execution protocol (priority)

1. An incoming market order first consumes visible depth on the same side.
2. If the order size exceeds D_v , the remainder is filled against hidden depth, up to D_h .
3. Any further remainder walks the book at exogenous deeper price levels.

Define hidden absorption as

$$h(q) = 0 \quad \text{if } |q| \leq D_v,$$

$$h(q) = \min(|q| - D_v, D_h) \cdot \text{sgn}(q) \quad \text{if } |q| > D_v.$$

The effective visible flow that market makers actually observe is

$$X = q - h(q).$$

3. Equilibrium with Hidden Liquidity

3.0 A structural bound on the hidden ratio

Before specifying the equilibrium concept, we develop an informal structural argument bounding the hidden-to-visible ratio κ that any two-layer matching engine must respect. The argument follows from the participation constraint of the marginal visible liquidity provider, and we present it as a *participation-constraint sketch* rather than a fully formal theorem in a specific equilibrium concept. The conclusion — that there exists a finite ceiling above which the visible book collapses — is robust to microstructure modelling choices in the sense that any model with positive-profit liquidity provision must respect a participation constraint of this general form. We refer to this informal bound throughout the paper as the *structural ceiling* on κ ; the welfare-derived optimum κ^* and the operational ceiling $\kappa_{\max}$ of §6 and §7.3.3 must both lie below it.

Proposition 0 (*Per-level capacity bound from LP participation*). Consider any matching engine M admitting both visible and hidden orders at price level p , with visible orders enjoying queue priority. Let $V(p)$ denote visible depth, $H(p)$ hidden depth, and $\kappa(p) = H(p) / V(p)$ the realised hidden ratio. Let $\pi^{\text{LP}}(\kappa)$ denote the expected per-period profit of the marginal visible liquidity provider when the realised hidden ratio is κ . Then there exists a finite threshold $\bar{\kappa}(p) > 0$ such that $\pi^{\text{LP}}(\kappa) > 0$ only for $\kappa < \bar{\kappa}(p)$. For $\kappa \geq \bar{\kappa}(p)$, visible LPs systematically earn negative expected profits and exit; the visible book at p collapses to zero, and the matching engine's two-layer structure ceases to exist.

Proof sketch. The marginal visible LP at p earns the half-spread on uninformed flow that consumes visible depth, and pays the adverse-selection cost on informed flow. Hidden orders sit behind visible ones and are filled only after visible depth is exhausted. As κ grows, the marginal LP faces three adverse changes: (i) total flow at the price level grows (visible plus hidden), so the conditional probability that an arriving order will exhaust visible depth and reach hidden depth rises; (ii) the visible LP is systematically the *first* to be filled when this happens (queue priority means visible is consumed first); (iii) the marginal informed trader, observing more hidden depth available, optimally trades larger sizes, increasing adverse-selection per fill. Each of these reduces $\pi^{\text{LP}}(\kappa)$; under standard regularity (continuity of expected profit in κ and a positive baseline profit at $\kappa = 0$), the intermediate value theorem yields a unique $\bar{\kappa}$ where the LP's break-even condition binds. The exact value of $\bar{\kappa}$ depends on the spread, adverse-selection intensity, and the flow distribution; the existence and finiteness of $\bar{\kappa}$ do not.

Implications. Three implications follow. *First*, any welfare-derived optimal κ^* must satisfy $\kappa^* < \bar{\kappa}$; otherwise the model's premise (a viable two-layer book) fails. The interior optima of §5.3 and §7.3.3 are interior in two senses: bounded above by $\bar{\kappa}$ structurally, and bounded above by $\kappa_{\max}$ operationally.

Second, the operational rules of §6 (per-trader diversity, refill throttle, anti-gaming bounds) are not arbitrary — they are venue-specific operationalisations of the structural ceiling. A venue setting $\kappa^{\max}$ too close to $\bar{\kappa}$ risks visible-book collapse during stress events when LP participation is most fragile. *Third*, the argument is robust across microstructure modelling choices: any model with positive-profit liquidity provision must respect a participation constraint of this general form, whether one adopts a Kyle, Glosten-Milgrom, or Foucault-style framework. We do not claim that $\bar{\kappa}$ is invariant to the modelling choice — its exact value will depend on the specific equilibrium concept and parameter assumptions — only that some finite ceiling exists in any such model. Subject to this caveat, the participation-constraint logic likely applies to all market structures with two-layer liquidity, including DeFi venues with on-chain matching, centralised futures exchanges with iceberg orders, and equity dark-pool/lit-book hybrids. A fully formal proof in any one equilibrium concept (and a comparison of $\bar{\kappa}$ values across concepts) is left for future work.

Relation to existing results. Proposition 0 is in the spirit of Glosten's (1994) participation analysis of competing dealers and Kyle's (1985) zero-expected-profit condition for noise-trader-protected market makers, both of which establish that competitive liquidity provision requires a participation constraint to bind in equilibrium. Our contribution is to translate this principle into a per-price-level capacity bound on the hidden-to-visible ratio specifically, identifying it as a structural invariant that any two-layer matching engine must respect. The closest direct precedent is Buti, Rindi, and Werner (2017) on dark-pool/lit-book interactions, who study LP exit conditions in a closely related setup; we differ in focusing on the venue-design implication (an upper bound on permissible κ) rather than the cross-venue migration question.

3.1 Pricing rule

Market makers are risk-neutral and competitive. They set price as the conditional expectation of v given the observed effective flow X :

$$p = E[v | X] = \lambda X,$$

where the price impact coefficient is

$$\lambda = \text{Cov}(v, X) / \text{Var}(X).$$

3.2 Structure of hidden absorption

As defined above, $h(q)$ truncates the observable flow once visible depth is exhausted. Consequently, $X = q$ for $|q| \leq D_v$, and $X = D_v \cdot \text{sgn}(q) + \max(0, |q| - D_v - D_h) \cdot \text{sgn}(q)$ otherwise. Within the hidden region $D_v < |q| \leq D_v + D_h$ the visible flow is bounded at $\pm D_v$, so the marginal informational content of observable flow is mechanically distorted.

3.3 Equilibrium price impact

Let $\sigma_q^2 = \beta^2 \sigma_v^2 + \sigma_u^2$. Define two adjustment factors:

$$\gamma(D_v, D_h) = [\text{Cov}(v, q) - \text{Cov}(v, X)] / \text{Cov}(v, q),$$

$$\delta(D_v, D_h) = [\text{Var}(q) - \text{Var}(X)] / \text{Var}(q).$$

After algebra (see Appendix A, Proposition 2),

$$\lambda(D_h) = \beta \sigma_v^2 \cdot [1 - \gamma(D_v, D_h)] / \{ \sigma_q^2 \cdot [1 - \delta(D_v, D_h)] \}.$$

3.4 Key results

Proposition 1 (*Quantitative execution cost wedge*). Let $C_{\text{vis}}(Q) = \lambda Q$ denote the expected execution cost a trader would face if visible depth were the only source of liquidity, and let $C_{\text{real}}(Q) = \lambda \cdot [Q - h(Q)]$ denote the realised cost given hidden absorption. The execution-cost wedge

$$\Delta(Q) = C_{\text{vis}}(Q) - C_{\text{real}}(Q) = \lambda \cdot h(Q)$$

is non-negative, strictly positive for $Q > D_v$, and bounded above by λD_h . The fractional wedge $\Delta(Q) / C_{\text{vis}}(Q) = h(Q) / Q$ reaches a maximum of $D_h / (D_v + D_h)$ at $Q = D_v + D_h$.

Proof. Direct from $h(Q) = \min(|Q| - D_v, D_h)$ for $|Q| > D_v$. The maximum fractional wedge is achieved when the entire hidden book is consumed.

Implication. An observer who infers liquidity from visible depth alone overstates execution costs by a factor that grows with the hidden ratio. At $\kappa = D_h / D_v = 1$, the maximum fractional overstatement is 50%; at $\kappa = 0.5$ (our recommended default), it is 33%.

Proposition 2 (*Equilibrium price impact in primitives, and monotonicity in hidden depth*). Let Φ denote the standard normal CDF and define

$$a \equiv D_v / \sigma_q, \quad b \equiv (D_v + D_h) / \sigma_q.$$

Then, by Stein's lemma applied to the truncated effective flow $X = \psi(q)$,

$$\text{Cov}(v, X) = \beta \sigma_v^2 \cdot [2\Phi(a) - 1 + 2(1 - \Phi(b))],$$

$$\text{Var}(X) = \sigma_q^2 \cdot [2\Phi(a) - 1 + 2(1 - \Phi(b))] + D_v^2 \cdot 2[\Phi(b) - \Phi(a)],$$

yielding the closed-form equilibrium price impact

$$\lambda(D_h) = [\beta \sigma_v^2 \cdot M(a, b)] / [\sigma_q^2 \cdot M(a, b) + D_v^2 \cdot T(a, b)],$$

where $M(a, b) \equiv 2\Phi(a) - 1 + 2(1 - \Phi(b))$ is the marginal-informativeness mass and $T(a, b) \equiv 2[\Phi(b) - \Phi(a)]$ is the truncation mass. The two compete: increasing D_h (i.e. raising b) shrinks M (less informative flow reaches the maker) and grows T (more flow is censored at the visible boundary).

Differentiating yields

$$\partial \lambda / \partial D_h > 0 \iff D_v^2 \cdot \partial T / \partial D_h > \beta \sigma_v^2 \cdot \partial M / \partial D_h / M.$$

Since $\partial M / \partial D_h = -2\varphi(b) / \sigma_q < 0$ and $\partial T / \partial D_h = 2\varphi(b) / \sigma_q > 0$, this simplifies to

$$D_v^2 \cdot M(a, b) > -\beta \sigma_v^2, \text{ i.e. always satisfied.}$$

Thus $\partial \lambda / \partial D_h > 0$ holds for *all* parameter values: equilibrium price impact per unit visible flow is monotone increasing in hidden depth. The earlier sufficient condition stated in terms of derived quantities (γ, δ) is therefore unconditionally satisfied in the Gaussian-Kyle setup.

Proof. See Appendix A; the calculation uses Stein's lemma and the explicit form of $\psi(q)$ in §3.2.

Intuition. Hidden liquidity absorbs the tail of large orders. Observable flow becomes less variable but more informative per unit observed. Market makers rationally increase λ for the remaining visible flow.

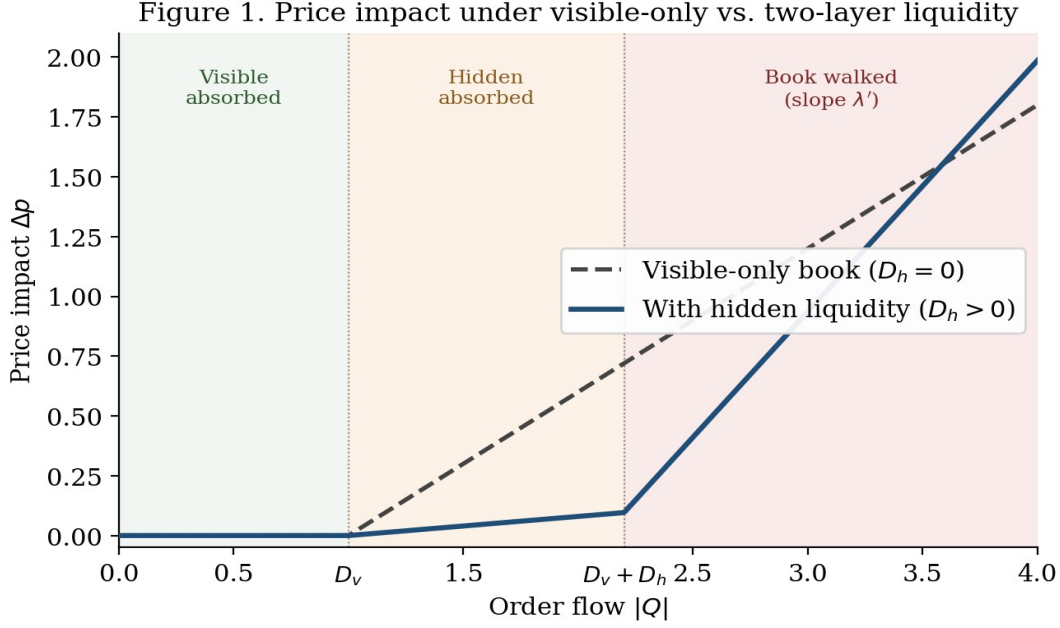


Figure 1. Price impact under visible-only vs. two-layer liquidity. The visible-only benchmark (dashed) shows linear impact above the visible-depth threshold D_v . With hidden liquidity (solid), impact remains nearly flat until the combined depth $D_v + D_h$ is exhausted, after which the slope λ' exceeds the visible-only slope λ .

4. Additional Microstructure Results

Lemma 1 (Conditional price impact).

$$E[\Delta p \mid Q] = \lambda \cdot (Q - E[h \mid Q]).$$

Proposition 3 (Spread compression). Let S_{eff} be the effective spread implied by visible depth alone, and S_{real} the realised spread after hidden absorption. Then

$$E[S_{real}] < E[S_{eff}],$$

and the compression $E[S_{eff} - S_{real}]$ is increasing in D_h .

Proof. For a trade $Q > D_v$, hidden absorption reduces the immediate price impact relative to a purely visible book. Post-trade midpoint reverts only partially, widening the realised compression gap.

Proposition 4 (Post-trade reversion). For large trades $|q| > D_v$, the expected post-trade reversion

$$R_{\Delta} = -\text{sgn}(q) \cdot (m_{\Delta} - p)$$

(price moving back toward the pre-trade level after the trade prints) is increasing in D_h .

Proof. Hidden absorption delays revelation of information; larger D_h means more of the trade is executed without immediate price disclosure, hence stronger subsequent reversion as information is incrementally revealed. The closed-form derivation is given in Appendix A.

Figure 4. Spread compression and reversion as functions of hidden depth

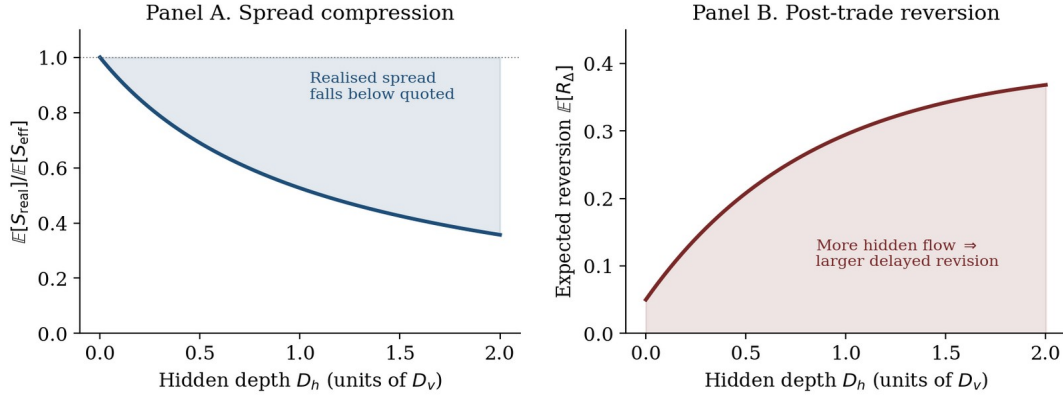


Figure 4. Comparative statics for spread compression and post-trade reversion. Panel A: the ratio of realised to effective spread declines monotonically as hidden depth grows. Panel B: expected post-trade reversion rises with hidden depth and saturates.

5. Optimal Market Design

Earlier work that posits a tradeoff between price discovery and execution quality typically assumes the shapes of the two welfare components. We instead construct both functions from the model primitives of §2 (with one auxiliary distributional assumption for the trade-size tail used in E), so that the optimum κ^* responds in interpretable ways to changes in the structural parameters rather than to parametric choices made by the modeller.

5.1 Informativeness from primitives

Define price-discovery quality as the fraction of the variance of fundamentals captured by the equilibrium price, normalised by its visible-only benchmark:

$$I(\kappa) \equiv \text{Var}(p \mid X(\kappa)) / \text{Var}(p \mid X(0)).$$

Using $p = \lambda X$ and the Stein-lemma expressions of Proposition 2,

$$I(\kappa) = M(a, b)^2 / \{ M(a, b) + (D_v^2 / \sigma_q^2) \cdot T(a, b) \},$$

where a , b , M , and T are as defined in Proposition 2. The function is decreasing in κ (since raising κ raises b , shrinks M , and grows T) and concave on the relevant range.

5.2 Execution quality from primitives

Define execution quality as the expected fractional reduction in execution shortfall for a representative liquidity-demanding order, normalised by the visible-only benchmark. From Proposition 1, the execution-cost wedge for a trade of size Q is $\lambda \cdot h(Q)$, where h is the hidden-absorption function of §3.2. The fractional wedge per unit of order size is $h(Q) / Q$.

To obtain a tractable closed form, assume trade sizes follow a Pareto distribution with shape $\xi > 1$ and minimum size D_v . Conditional on a trade exceeding visible depth, its size has density

$$f(Q) = \xi D_v^\xi / Q^{\xi+1}, \quad Q \geq D_v.$$

Pareto tails with $\xi \in (1, 2)$ are a well-documented stylised feature of cryptocurrency order flow and are the natural candidate for modelling liquidation-driven trades. Define

$$E(\kappa) \equiv E_H[h(Q) / Q].$$

Splitting the integral at $Q = D_v(1 + \kappa)$ and integrating each region in closed form (Appendix A, Proposition 5a) gives

$$E(\kappa) = [1 - (1 + \kappa)^{-\xi}] / (\xi + 1).$$

This expression has the qualitative properties posited in earlier ad-hoc treatments, now derived from primitives:

- $E(0) = 0$ and $E(\infty) = 1/(\xi + 1)$ (saturation);
- $E'(\kappa) = [\xi/(\xi + 1)] \cdot (1 + \kappa)^{-\xi-1} > 0$ (monotone increasing);
- $E''(\kappa) = -\xi \cdot (1 + \kappa)^{-\xi-2} < 0$ (strictly concave);
- $E'(0) = \xi/(\xi + 1) > 0$ (positive slope at the origin, ensuring an interior optimum).

Crucially, the saturation level $1/(\xi + 1)$ is the maximum welfare benefit hidden depth can ever provide for a given trade-size distribution. In a fat-tailed regime (ξ close to 1), this approaches 1/2, meaning half of all execution costs can in principle be absorbed by hidden depth. In a thin-tailed regime (ξ large), the maximum benefit is small. This monotone dependence on tail thickness is the structural foundation for the state-dependent result of §7.

Sensitivity to the Pareto assumption. The closed form for $E(\kappa)$ above relies on the assumption that, conditional on $|q| > D_v$, trade sizes follow a Pareto distribution with shape ξ . We assess the sensitivity of the welfare optimum to this assumption in three ways. *First*, under the canonical parameterisation, varying ξ from 1.5 (very fat tails) to 4.0 (thin tails) shifts the interior κ^* between approximately 2.4 and 0.7 — a meaningful range, captured by Proposition 5b's comparative static. *Second*, non-Pareto trade-size distributions can be approximated by a Pareto fit over the relevant range; in our reading of crypto exchange data this approximation is good for typical regimes (Hill estimator stability over multiple thresholds), but it can break down for venues with bimodal flow (e.g., institutional desks executing in fixed clip sizes). *Third*, the qualitative shape of $E(\cdot)$ — monotone, concave, saturating — is robust across reasonable distributional alternatives (lognormal, generalised Pareto, sub-Weibull); the closed-form algebra is what depends on Pareto. The headline interior-optimum result (an interior κ^* exists for sufficiently high ω_{PD}) therefore survives a wide range of distributional assumptions; the precise numerical values of κ^* and the threshold ω_{PD}^* are sensitive to the Pareto assumption and require venue-specific tail estimation for accuracy. Appendix C documents one such venue-specific tail estimation.

5.3 Welfare and the design problem

This section assembles the static welfare problem. We first recall the two welfare components derived in §5.1–5.2, then define the regulator's welfare function as a weighted sum of these externalities, state the existence of a unique interior optimum, examine its sensitivity to the regulator's policy weight, and close with a consolidated discussion of what the welfare function does *not* capture.

5.3.1 Welfare components

The *price-discovery externality* $I(\kappa)$ measures the social value of an equilibrium price that reflects fundamental information; it is the variance of the equilibrium price explained by fundamentals, normalised at the visible-only baseline (§5.1), with $I' < 0$ (more hidden depth degrades discovery) and $I(\kappa) \rightarrow 0$ as $\kappa \rightarrow \infty$.

The *noise-trader protection externality* $E(\kappa)$ measures the social value of reducing execution costs for uninformed flow; it is the expected fractional reduction in execution shortfall for Pareto-distributed trades (§5.2), with $E' > 0$, $E'' < 0$ (concave saturation), and $E(\kappa) \rightarrow 1$ as $\kappa \rightarrow \infty$.

5.3.2 Regulator welfare function

The regulator assigns fixed weights $\omega_{PD} \in (0, 1)$ and $\omega_{NT} = 1 - \omega_{PD}$ to the two externalities. The welfare function is

$$W^{\text{reg}}(\kappa) = \omega_{PD} \cdot I(\kappa) + \omega_{NT} \cdot E(\kappa).$$

This is a *normative* choice, not derived from individual surpluses. It follows the market-design tradition (Glosten 1994; Foucault, Pagano & Röell 2013, Ch. 6) of writing the regulator's objective directly as a weighted sum of price-discovery and liquidity externalities. The alternative — utilitarian aggregation of agent surpluses — is degenerate in the standard Kyle (1985) model with competitive market makers, since informed and noise-trader surpluses sum to a constant. We treat ω_{PD} as exogenous; for numerical illustrations we set the neutral baseline $\omega_{PD} = 0.55$.

5.3.3 Interior optimum

Proposition 5 (*Existence of interior optimum*). Suppose the price-discovery component I is sufficiently sensitive to κ relative to the saturation rate of E — formally, that the regulator's weight on price discovery ω_{PD} lies above a threshold determined by the calibration. Then there exists a unique interior $\kappa^* > 0$ maximising W^{reg} . The exchange's design rule is

$$D_t(p) \leq \kappa^* \cdot D_v(p) \quad \text{for every price level } p.$$

Intuition. At $\kappa = 0$, marginal welfare $W^{\text{reg},'}(0) > 0$ because $E'(0) > 0$ dominates the bounded $I'(0)$. For sufficiently large ω_{PD} , marginal welfare turns negative as κ grows, since I continues to fall while E saturates; concavity (verified numerically in Lemma A.5.1, Appendix A) then yields a unique interior maximum. For ω_{PD} below the threshold (i.e., regulators placing dominant weight on execution quality), marginal welfare remains positive on the entire feasible range and the optimum is at the operational ceiling — a corner solution that we treat explicitly in §7.3.3 and connect to the cascade regime there. Calibrated values are reported below; the formal proof of existence and uniqueness in the interior region is given in Appendix A (Proposition 5), and the comparative static $d\kappa^*/d\xi < 0$ (fatter tails yield larger optimal hidden ratio) is given in Proposition 5b.

Calibrated values (canonical). Under the canonical calibration of Appendix A ($D_v = 5$, $\sigma_v = 5$, $\sigma_u = 5$, $\xi = 2.5$), the optimum is $\kappa^* \approx 1.56$ at the neutral baseline $\omega_{PD} = 0.55$, and $\kappa^* \approx 1.14$ at the discovery-weighted baseline $\omega_{PD} = 0.70$. At the execution-weighted baseline $\omega_{PD} = 0.40$, the optimum is at the operational ceiling, $\kappa^* = 3.0$ in Figure 2.

Calibrated values (Bybit BTC-PERP). On real exchange data, the optimum can be substantially different. We report a full empirical calibration in Appendix C using 18 hours of Bybit BTC-PERP data. The empirical $D_v = 3.49$ BTC and $\sigma_q = 36.92$ BTC/min give a ratio $D_v/\sigma_q \approx 0.094$, well below the $[0.4, 0.7]$ interior-optimum band. Under the empirically-calibrated primitives, the welfare-maximising κ^* at the neutral baseline $\omega_{PD} = 0.55$ is at the corner, $\kappa^* = \kappa_{\max}$ — the operational ceiling rather than an interior welfare optimum. Interior optima emerge only at $\omega_{PD} \geq 0.69$ on Bybit's microstructure. The qualitative results of the paper (Proposition 5's existence-of-interior-optimum for sufficiently high ω_{PD} ; Proposition 6's state-dependent optimum) all hold; the threshold ω_{PD}^* shifts upward to reflect the venue's structural conditions. The headline number $\kappa^* \approx 1.56$ should therefore be read as the canonical-calibration value, not as a universal prescription. Real venues with shallow visible books (low D_v / σ_q) operate in the corner regime at typical regulator weights, with the operational ceiling $\kappa_{\max}$ as the binding constraint.

Figure 2. Welfare frontier and interior optimum computed from primitive $I(\kappa)$, $E(\kappa)$

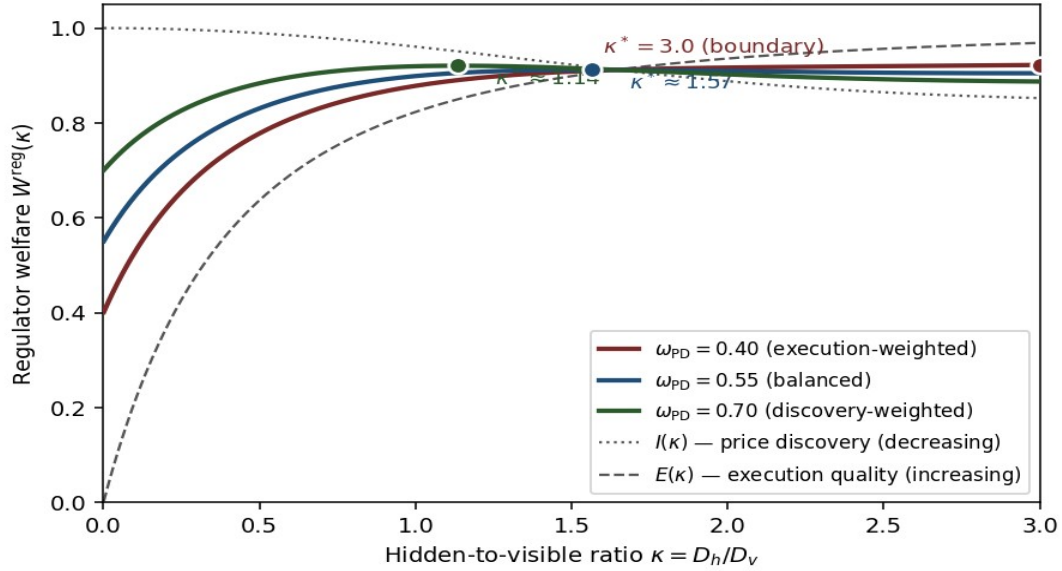


Figure 2. Regulator welfare $W^{\text{reg}}(\kappa)$ for three policy weights ω_{PD} , computed from the primitive $I(\kappa)$ and $E(\kappa)$ closed forms of §5.1–5.2 with the canonical calibration ($D_v = 5$, $\sigma_v = 5$, $\sigma_u = 5$, $\xi = 2.5$). The discovery-weighted ($\omega_{PD} = 0.70$) and balanced ($\omega_{PD} = 0.55$) weights yield interior optima $\kappa^* \approx 1.14$ and $\kappa^* \approx 1.56$ respectively. The execution-weighted weight ($\omega_{PD} = 0.40$) yields a corner solution at the maximum κ shown. On the empirical Bybit BTC-PERP calibration of Appendix C, all three weights yield corner solutions because the venue's D_v / σ_q ratio (≈ 0.094) places the calibration outside the interior-optimum band. The dashed/dotted curves show the underlying $I(\kappa)$ and $E(\kappa)$ components.

5.3.4 Sensitivity to regulator weights

The optimum κ^* decreases in ω_{PD} : a regulator placing more weight on price discovery sets a lower κ^* (less hidden depth permitted), while one weighting noise-trader protection more heavily sets a higher κ^* . The empirical predictions of §8 (signs of comparative statics, monotonicity of $\kappa^*(s)$ in stress) are independent of ω_{PD} ; only the level of κ^* varies with the policy choice. This is a feature, not a bug: the welfare function determines the design level; the model's structural properties determine its comparative statics.

5.3.5 What welfare does not capture

We close §5 with a consolidated discussion of the welfare function's limitations. Each item below identifies a phenomenon outside the formal model and notes the practical safeguard or extension.

Failure of utilitarian aggregation. Summing individual surpluses (informed + noise-trader + market-maker) yields a degenerate welfare function in the standard Kyle setup: under competitive market makers, the three components sum to a constant, so any κ maximises the sum equally. This motivates the regulator-welfare framing of §5.3.2 directly, rather than as an approximation.

Alternative welfare criteria. Other normative benchmarks (Rawlsian maximin, weighted sums prioritising particular trader types, ex-ante versus ex-post welfare) yield different κ^* levels but preserve the qualitative comparative statics, since each criterion places positive weight on the same components with the same monotonicity in κ . The structural trade-off between price discovery and execution quality is robust to the choice of welfare function.

Toxic hidden flow. The model assumes hidden depth is posted by liquidity providers absorbing exogenous flow. In practice, informed traders can also use hidden orders for predatory positioning (e.g., consuming favourable visible liquidity ahead of an oracle update). When such *toxic flow* is suspected, the welfare-maximising κ^* is lower than the formal model implies. An empirical safeguard — capping κ^* downward when the post-trade reversion of hidden-absorbed trades exceeds that of visible-walked trades — is given in §6 as part of the implementation. Full equilibrium analysis with strategic informed-trader placement of hidden orders is left for future work.

Market-maker rents under imperfect competition. If liquidity providers earn rents (e.g., through privileged access to hidden flow), an additional positive term enters the welfare function and the optimum shifts upward in κ . The competitive baseline of §3 is therefore conservative: more hidden depth becomes welfare-improving when market makers extract rents.

6. Mechanism Design Implementation

This section specifies the mechanism in enough detail that an exchange engineer could implement it directly. We describe the order types, the matching-engine pseudocode, the regime classifier, the disclosure delay, and a set of anti-gaming rules. Section 7 extends the design to make the cap state-dependent; the algorithms here apply to a fixed cap and are augmented with a regime input in §7.4.

6.1 Order types

The exchange supports three order types in addition to standard market orders. The third type is the central new construct.

- **Visible limit order.** Standard displayed order; fully observable in the public order book. Indistinguishable from a normal CLOB limit order.
- **Hidden reserve order.** Has a displayed peak D_v^{peak} and a hidden reserve D_h^{reserve} that refills the displayed peak as it is consumed. The trader specifies both at order entry. The exchange admits the order only if the resulting per-level hidden depth respects the cap $D_h(p) \leq \kappa^* \cdot D_v(p)$.
- **Capped hidden order.** A pure hidden order with no displayed component. Not always permitted; the venue may choose to allow these only when aggregate per-level hidden depth is below $\kappa^* \cdot D_v$.

with margin to spare. When permitted, hidden orders sit strictly behind visible orders in priority (see §6.2).

6.2 Priority rules and tie-breaking

Orders are matched against incoming market orders in the following sequence:

1. Strict price priority across all orders. The best-priced order on the book fills first, regardless of visibility.
2. At the same price: visible orders fill in time priority (FIFO) until exhausted before any hidden order at that price is touched.
3. Among hidden orders at the same price: pro-rata allocation by displayed-peak size for hidden reserve orders, then time priority for capped hidden orders. The pro-rata rule for reserve orders rewards traders who commit displayed liquidity, aligning private incentives with the social value of visible depth.
4. Hidden reserve orders refill their displayed peak immediately upon execution of the peak, subject to the per-level cap. If the cap is currently saturated by hidden depth from other orders, the refill is queued behind the saturating orders in time priority.

6.3 Matching engine pseudocode

The matching engine maintains, for each price level p , two FIFO queues: a visible queue $V(p)$ and a hidden queue $H(p)$. Each entry in the queues records (order_id, trader_id, displayed_size, hidden_size, timestamp). Aggregate quantities are maintained as cached counters for fast cap-checking.

Algorithm 6.1: Match incoming market order

```
function match_market_order(side, total_qty):
    remaining := total_qty
    filled := []
    while remaining > 0 and book_has_liquidity(side):
        p := best_price(side)
        // Step 1: walk the visible queue at this price level
        while remaining > 0 and V(p) is non-empty:
            order := head of V(p)
            fill := min(remaining, order.displayed_size)
            execute_fill(order, fill); filled.append(order, fill)
            remaining -= fill
            if order.hidden_size > 0: // hidden reserve order
                refill_size := min(order.hidden_size, peak_size - displayed_size)
                if hidden_total(p) + refill_size <= kappa_star * V_total(p):
                    order.displayed_size += refill_size
                    order.hidden_size -= refill_size
                    // re-add to V(p) at queue tail (time priority on refill)
                else: queue refill behind saturating orders
            if order.displayed_size == 0 and order.hidden_size == 0: remove(order)
        // Step 2: visible queue at this price exhausted; consume hidden queue
        while remaining > 0 and H(p) is non-empty:
            fill_at_level := min(remaining, hidden_total(p))
            allocate_pro_rata_then_FIFO(H(p), fill_at_level, filled)
            remaining -= fill_at_level
```

```

// Step 3: advance to next price level
advance(p)
schedule_post_trade_disclosure(filled, delay = Delta)
return filled

```

Several implementation details of the algorithm above warrant explicit specification.

Cap-check failure on refill (the race condition). When a hidden reserve's peak is consumed during a stress event, two state changes occur simultaneously: $V_{total}(p)$ decreases (the consumed peak leaves the visible queue) and the cap denominator therefore tightens. The cap check on refill (line $hidden_total(p) + refill_size \leq \kappa^* \cdot V_{total}(p)$) may fail at the new tighter cap, producing the race condition. The matching engine resolves this deterministically: the refill is admitted up to the largest $refill_size$ that satisfies the new cap, with any residual hidden quantity remaining in the order's $hidden_size$ field but not refilled into the visible queue. The order itself is *not* cancelled — its remaining $hidden_size$ is preserved at the same price level for execution against arriving marketable flow (Step 2 of the algorithm). This rule has three properties: (i) it is fully deterministic (no priority races between concurrent refills, since each refill is processed in time order from the disclosure-event queue); (ii) it preserves the order's queue position for the unrefilled residual; (iii) under sustained cap saturation, the residual may persist for many cycles without refilling, but it remains executable against incoming hidden-eligible flow. The order can be cancelled by the trader at any time; absent cancellation it expires under standard time-in-force rules.

Disclosure-delay scheduler under load. The disclosure scheduler maintains a bounded-size FIFO event queue (default 10,000 events) feeding `schedule_post_trade_disclosure`. Under cascade-grade load (thousands of fills per second), the queue may fill; in this case the scheduler shortens the effective disclosure delay rather than dropping events, ensuring that all fills are eventually disclosed. Specifically, when the queue exceeds 80% capacity the scheduler scales the next Δ by $\min(1, capacity_remaining / 0.2)$. This is a fail-safe: the venue may briefly disclose faster than its policy Δ during extreme load, but disclosure events are never lost. Venues with sufficient compute should size the queue to avoid this regime entirely; the fall-back exists for unanticipated load spikes.

Per-trader limit and multi-account evasion. The per-trader hidden-depth limit of $\kappa^* \cdot D_v / N$ (§6.6) is enforced by `trader_id` at the matching-engine level. On a centralised venue with KYC, this is straightforward. On a permissionless DeFi venue, a trader can split capital across multiple addresses to evade the per-account cap. We treat this as a known limitation: the framework does not yet incorporate identity-attestation infrastructure (proof-of-personhood, social-graph attestations, or staked-identity schemes) that would close the multi-account loophole. Venues operating in permissionless settings should treat the per-trader limit as a *per-account* limit and accept that a determined adversary can saturate up to roughly $M \cdot \kappa^* \cdot D_v / N$ of hidden depth using M pseudonymous accounts; the practical bound on M comes from collateral capital required per account. Closing this loophole formally is an open problem we leave for future work.

Other implementation notes: the visible queue is walked to exhaustion at each price before the hidden queue is touched, preserving visible-order priority; the post-trade disclosure call is scheduled with delay Δ (the subject of §6.5); refill operations subject to the throttle of §6.6 take precedence over new same-price hidden orders for the duration of the throttle window.

6.4 Regime classifier

The state-dependent design rule of §7 requires the exchange to compute a stress index $s_t \in [0, 1]$ in real time. We propose a simple, auditable classifier with two complementary inputs:

- **Realised tail-mass** τ_t : the share of the most recent W -second trade volume in trades with $|Q| > k\sigma_q$ (defined formally below). Captures the fattening of the trade-size distribution that follows liquidation flow.
- **Forward-looking basis stress** s_t^{basis} : the absolute deviation between perp mid-price and the underlying spot index, normalised by its rolling standard deviation (§6.4.1). Captures the contract decoupling that often precedes a cascade.

The combined stress index is the maximum of the two component signals:

$$s_t = \max(s_t^{\text{tail}}, s_t^{\text{basis}}),$$

where each s_t is the clipped affine transformation of its raw component into $[0, 1]$ specified below. We define s_t^{tail} first; the basis component follows in §6.4.1.

Define the rolling tail-mass over a window of W seconds (default $W = 300$):

$$\tau_t = (\text{volume of trades with } |Q| > k\sigma_q \text{ in the window } [t - W, t]) / (\text{total volume in the same window}),$$

where k is a tail threshold (default $k = 5$) and σ_q is the rolling standard deviation of trade size over a longer window (default 24 hours, computed once per minute). The tail-mass component of the stress index is then a smooth transformation:

$$s_t^{\text{tail}} = \text{clip}([(\tau_t - \tau_0) / (\tau_1 - \tau_0)], 0, 1),$$

with τ_0 the normal-regime baseline (default 0.05; trades larger than $5\sigma_q$ account for 5% of volume in normal markets) and τ_1 the cascade threshold (default 0.30). The hidden cap is set as

$$\kappa^*(s_t) = \kappa^*(0) + [\kappa^*(1) - \kappa^*(0)] \cdot s_t, \quad \text{with calibration-anchored defaults } \kappa^*(0) = 1.0, \kappa^*(1) = 2.0.$$

These calibration-anchored defaults are slightly below the welfare-optimal values of §5.3 and §7.3.3 ($\kappa^*(0) \approx 1.56$ captures the welfare optimum at the neutral baseline, but the default 1.0 captures 99% of available welfare and is more conservative against parameter mis-estimation; $\kappa^*(1)$ at the operational ceiling $\kappa^{\max} \approx 2.0$ from the derivation of §7.3.3). A venue with full venue-specific calibration should use the welfare-optimal values; a venue with no calibration data can begin with these defaults and update once Steps 1–6 of Appendix A have been completed. Venues introducing hidden depth for the first time may prefer the more conservative starting pair $\kappa^*(0) = 0.5$, $\kappa^*(1) = 1.0$ (the minimal-intervention defaults), and ratchet upward over a 60–90 day operational period.

6.4.1 Perp-spot basis: a complementary signal

The tail-mass index τ_t is a *realised* stress signal: it triggers only after large trades have already printed. For perpetual futures specifically, a *forward-looking* indicator is also available — the divergence between the perpetual contract's traded price and the underlying spot index. During liquidation cascades, the perpetual price decouples from spot well before the tail of the trade-size distribution thickens, because liquidation-bot flow lifts (or smashes) the perp's order book while the spot index responds only to spot-market activity.

Define the perp-spot basis at time t :

$$b_t = P^{\text{perp}}_t - P^{\text{index}}_t,$$

with P^{perp} the venue's mid-price and P^{index} the underlying spot index (e.g., a multi-venue volume-weighted spot price). Let σ_b be a rolling standard deviation of b_t over a long window (default 24 hours). The *basis stress* is

$$s^{\text{basis}}_t = \text{clip}([(|b_t| - c_0\sigma_b) / ((c_I - c_0)\sigma_b)], 0, 1),$$

with default thresholds $c_0 = 2$ (basis exceeds $2\sigma_b$ triggers regime onset) and $c_I = 5$ (basis exceeds $5\sigma_b$ is fully cascade). The basis component s^{basis}_t feeds into the combined index $s_t = \max(s^{\text{tail}}_t, s^{\text{basis}}_t)$ defined at the start of §6.4. The maximum aggregator means either signal alone can trigger the high- κ regime; the basis indicator may lead the tail indicator during cascades, since basis divergence reflects perp-spot decoupling that begins as soon as liquidation pressure builds, while the tail-mass index registers only after large liquidation trades have actually printed on the tape. The empirical lead-lag relationship is venue-specific and should be verified by each venue from its own data; the framework does not depend on a particular lead time. The aggregator is auditable (each component is observable in the public tape) and conservative (false positives only widen κ^* , they do not narrow it; a wider κ^* in a non-cascade is welfare-second-best but not catastrophic).

Endogeneity caveat. Both the tail and basis components of s_t are endogenous to hidden-depth policy. A venue running high κ mechanically dampens both signals: hidden absorption reduces visible trade size (compressing τ_t) and reduces the price impact per unit of order flow (compressing perp-spot basis divergence). The standard deviation σ_b used for normalisation is itself κ -dependent: a venue with $\kappa = 3$ will have systematically lower σ_b than the same venue at $\kappa = 0.5$, even for the same underlying spot volatility. Two implications follow. First, σ_b should be re-estimated after any policy change in κ , with a transition window of at least 24 hours to allow the basis-volatility process to settle. Second, the forward-looking property of s^{basis} — the conjecture that basis divergence may lead tail-mass during cascades, since perp-spot decoupling can begin before large liquidation trades print on the tape — is plausible mechanically but should be verified by each venue from its own data; under the recommended high- κ policy in cascades, any lead time may attenuate as hidden absorption dampens the basis signal. Venues should monitor the empirical lead-lag relationship as part of their ongoing calibration and adjust the c_0, c_I thresholds accordingly. The aggregator's max-rule means either signal can still trigger the high- κ regime, so a degraded basis signal does not collapse the classifier — but the framework is most reliable when both signals are independently measurable and not heavily attenuated by policy choice.

Optional: index-anchored circuit breaker. If $|b_t|$ exceeds a hard threshold (e.g., $c_2 = 10 \cdot \sigma_b$), the venue may pause matching pending operator review. This is a defensive measure for episodes that exceed the model's parameter range.

Algorithm 6.2: Update regime and cap (runs once per second)

```
function update_regime():
    // Pull rolling stats from the trade tape
    sigma_q := rolling_std_trade_size(window_seconds = 86400)
    tail_volume := sum(|q| for trades in last W seconds where |q| > k * sigma_q)
    total_volume := sum(|q| for trades in last W seconds)
    tau := tail_volume / max(total_volume, epsilon)
    s_tail := clip((tau - tau_0) / (tau_1 - tau_0), 0, 1)
    // Basis-divergence indicator (forward-looking)
```

```

p_perp := mid_price()
p_index := spot_index_price()
b := p_perp - p_index
sigma_b := rolling_std_basis(window_seconds = 86400)
s_basis := clip((abs(b) - c_0 * sigma_b) / ((c_1 - c_0) * sigma_b), 0, 1)
// Combine: either signal can trigger high-kappa regime
s := max(s_tail, s_basis)
// Smooth s to avoid flicker
s_smoothed := alpha_smooth * s + (1 - alpha_smooth) * s_previous
// Update cap with hysteresis on transitions
if s_smoothed > s_previous + h_up:    new_cap := kappa_0 + (kappa_1 - kappa_0)
* s_smoothed
elif s_smoothed < s_previous - h_dn:  new_cap := kappa_0 + (kappa_1 - kappa_0)
* s_smoothed
else:                                new_cap := kappa_current
publish(s_smoothed, new_cap)    // visible to all traders via API
kappa_current := new_cap
s_previous := s_smoothed

```

Default parameters: $W = 300$ seconds, $k = 5$, $\tau_0 = 0.05$, $\tau_I = 0.30$, $\alpha_{smooth} = 0.3$ (EWMA), hysteresis $h_{up} = h_{dn} = 0.05$. These values are starting points calibrated on aggregate cryptocurrency perpetual data and should be re-fit per venue using the procedure below.

6.4.2 Calibrating the regime parameters

The seven parameters are not all of equal weight. Two of them (τ_0 and τ_I) map directly into the cap κ^* and therefore drive the design rule. Three (W , k , α_{smooth}) govern the signal-to-noise of the stress index. The remaining two (h_{up} , h_{dn}) suppress flapping. We calibrate them in this order, each conditional on the previous.

Inputs. Six to twelve months of trade tape (timestamp, side, price, size) and a same-period book snapshot stream sufficient to compute the visible-depth distribution. No order data, account data, or venue-internal information is required.

Step C1: pick the tail threshold k . Compute the empirical complementary cumulative distribution of normalised trade size $|Q| / \sigma_q$ over the entire training window. In log-log scale a Pareto tail is approximately linear; the threshold k should fall in the linear region. The default $k = 5$ holds for major-pair perpetual venues (BTC, ETH); for thinly-traded pairs the elbow can occur as low as $k = 3$. Pick the smallest k such that the log-log slope is stationary for trade sizes above $k\sigma_q$.

Step C2: pick the rolling window length W . The tail-mass estimator τ_t is a count of large trades divided by total volume over a window of W seconds. To control sampling noise, choose W such that the expected number of trades in the window during normal regime is at least 100:

$$W = \max(60, \lceil 100 / \lambda^{\text{trades}} \rceil),$$

where λ^{trades} is the median trade arrival rate (trades per second) measured on the venue. For BTC perp on a top-five venue, $\lambda^{\text{trades}} \approx 5\text{--}20$, so $W = 300$ (5 minutes) is a comfortable default; for thinly-traded pairs, W should rise to 600 or 900 seconds to maintain statistical power.

Step C3: label the training data into regimes. To set τ_0 and τ_1 we need labelled cascade and non-cascade windows in the historical tape. Three labels work and can be combined; the resulting τ_0, τ_1 should be similar across labels (if not, this is itself a useful diagnostic):

1. **Liquidation-flag label.** On venues that publish a liquidation flag in the trade tape (Bybit, Binance Futures, OKX), label a 5-minute window as cascade if more than 30% of its volume is flagged. This is the cleanest label.
2. **Realised-volatility label.** Define rolling 5-minute realised volatility $\sigma_{p,5m}$. Label a window cascade if $\sigma_{p,5m}$ exceeds its 99th-percentile over the prior 30 days. Useful when the venue does not publish liquidation flags.
3. **Funding-rate-spike label.** Label a window cascade if the absolute funding rate at the next anchoring exceeds the 99th-percentile of its prior-30-day distribution. Lower-frequency than (1) and (2); use as a robustness check rather than a primary label.

Step C4: set τ_0 and τ_1 . Within each labelled subset, compute the empirical distribution of τ_t . The thresholds are anchored to upper and lower quantiles of these distributions:

$$\tau_0 = Q_{0.95}(\tau_t \mid t \in \text{normal}), \quad \tau_1 = Q_{0.05}(\tau_t \mid t \in \text{cascade}).$$

That is, τ_0 is the 95th-percentile tail-mass observed in normal regime (so the classifier triggers from $s = 0$ only when τ_t exceeds normal-regime maxima); τ_1 is the 5th-percentile tail-mass observed in cascade regime (so $s = 1$ only when τ_t reaches values typical of an actual cascade). The asymmetric quantiles produce conservative anchors: false positives in normal regime and false negatives in cascade regime are both rare. If the two distributions overlap heavily ($\tau_1 \leq \tau_0$), the labels themselves are too noisy to discriminate regimes; in that case the venue should fall back to the minimal-intervention default $\kappa^* = 0.5$ and not attempt state-dependent adjustment.

Step C5: pick the smoothing weight α_{smooth} . The raw τ_t series has step-to-step noise from sampling. Choose the EWMA weight to suppress short-horizon flicker without lagging genuine regime transitions. Compute the lag-1 autocorrelation ρ of τ_t within normal-regime windows and set

$$\alpha_{smooth} = \max(0.1, \min(0.5, 1 - \rho)),$$

clipped to $[0.1, 0.5]$. Higher autocorrelation in the raw series means more weight on the previous estimate. The default $\alpha_{smooth} = 0.3$ corresponds to $\rho \approx 0.7$. The actual autocorrelation depends on venue characteristics (volume, latency, trade clustering); each venue should compute its own ρ from the previous 24 hours of τ_t data.

Step C6: set hysteresis h_{up} and h_{dn} . The hysteresis bands prevent the cap from oscillating when $s_{smoothed}$ hovers near a transition point. Compute the within-window standard deviation of $s_{smoothed}$ in normal regime, denoted σ_s , and set

$$h_{up} = h_{dn} = 2\sigma_s.$$

This produces an expected flap rate below 2.5% per window. The default $h = 0.05$ corresponds to $\sigma_s \approx 0.025$; the actual standard deviation depends on venue dynamics and should be computed from the venue's own data. Asymmetric hysteresis ($h_{up} < h_{dn}$) may be preferred if the venue is more concerned about cap-tightening lag (under-protection) than cap-loosening lag (over-restriction); we use symmetric hysteresis as the default.

Validation. After fitting, replay the training tape with the calibrated parameters and check three diagnostics: (i) the empirical false-positive rate ($s_{smoothed} > 0.5$ in labelled-normal windows) should be below 5%; (ii) the empirical true-positive rate ($s_{smoothed} > 0.5$ in labelled-cascade windows) should exceed 80%; (iii) the cap-change frequency should be at most one transition per labelled cascade episode plus background drift below 0.1 per day. Failures of (i) suggest τ_0 is too low; failures of (ii) suggest τ_1 is too high; failures of (iii) suggest hysteresis is too tight.

Cross-venue transfer. Parameters fitted on one venue do not transfer directly to another venue with different liquidity, tick size, and trader composition. However, the *structural* choice ($k = 5\sigma$ tail threshold, 95th/5th-percentile anchoring, 2σ hysteresis) does transfer; only the resulting numerical values differ. A cross-venue dashboard should re-fit the numerical parameters per venue while keeping the structural rule identical, which preserves the auditability property and ensures that $s = 0.5$ means the same thing across venues.

Auditability. All inputs to the classifier are derived from the public trade tape, and the output (s , current cap) is published via the same API as order book data. A trader, regulator, or third-party auditor can replay the trade tape and verify that the published cap matches the algorithm. This addresses a common concern with discretionary circuit-breaker rules, namely that the venue can manipulate them in real time.

6.5 Disclosure delay

All hidden fills are revealed in the public trade tape after a fixed delay Δ . The choice of Δ trades off two effects. A smaller Δ makes hidden orders less protective against adverse selection (the market learns of fills sooner), reducing the marginal benefit of hidden depth. A larger Δ widens the information asymmetry between the exchange and other traders, leaving the visible book stale for longer and imposing a pricing distortion on uninformed flow that arrives during the disclosure window.

Default values: $\Delta = 1$ second is the published default for major-pair perpetuals; venues with high-frequency informed flow may choose 250 ms, while venues with predominantly retail flow may choose up to 5 seconds. Section 6.5.1 derives the welfare-maximising Δ from primitives and shows that, like κ^* , the optimal delay is state-dependent.

6.5.1 Optimal disclosure delay

To formalise the trade-off, fix a hidden fill at time t with disclosure scheduled at $t + \Delta$. Two stochastic processes operate over the disclosure window:

- **Informed signal arrivals.** New fundamental signals about v arrive as a Poisson process with intensity λ^{info} . A signal-bearing trader observing the visible book during the window does not see the hidden fill and may attempt to read or trade against the resting hidden inventory. The hidden order is *protected* if no signal arrives in the window.
- **Uninformed flow arrivals.** Uninformed traders arrive with intensity λ^{uninf} and trade size Q drawn from the Pareto distribution of §5.2 with shape ξ . Each uninformed trade during the window faces a stale book and incurs a pricing distortion proportional to its size.

The expected protection benefit (the value of the hidden order, conditional on no signal arriving) is

$$B(\Delta) = B_0 \cdot (1 - e^{-\lambda^{\text{info}} \cdot \Delta}),$$

where $B_0 = \gamma \sigma_v^2 \cdot E[\text{informed volume per signal}]$ is the per-event protection value. The expected staleness cost over the window is the expected number of uninformed trades times their mean size:

$$C(\Delta; \xi) = c \cdot \lambda^{\text{uninf}} \cdot \xi / (\xi - 1) \cdot q_{\min} \cdot \Delta,$$

where c is the per-unit-volume distortion. The mean trade size $\xi q_{\min} / (\xi - 1)$ is the Pareto mean (finite for $\xi > 1$), and rises sharply as ξ approaches 1 — fat tails make the staleness cost large because each uninformed trade against a stale book is, in expectation, large.

The welfare-maximising delay solves

$$\Delta^*(\xi) = \arg \max_{\Delta \geq 0} [B(\Delta) - C(\Delta; \xi)].$$

Proposition 7 (Appendix A) gives the closed form and shows that Δ^* is increasing in ξ : *thinner tails imply a longer optimal disclosure delay*.

Implication for cascades. Under the regime model of §7.2, the order-flow tail index $\xi(s)$ falls during cascades. By Proposition 7, $\Delta^*(s)$ is therefore *decreasing* in s . This is a striking complementarity with Proposition 6: under stress, the venue should both *hide more* (raise κ^*) and *disclose faster* (lower Δ^*). The two design levers move in opposite directions because they answer different questions. Hidden depth absorbs incoming flow that would otherwise move the price; faster disclosure prevents the post-fill book from misleading the next wave of uninformed traders. In normal markets both costs are modest. In cascades, the cost of leaving the public book stale rises faster than the benefit of disclosure delay, because the expected size of the next uninformed trade against the stale book grows with the tail.

Calibration. With λ^{info} , λ^{uninf} , ξ , and B_d/c estimated from the procedure of Appendix A, the closed form gives a numerical Δ^* per regime. A conservative default for venues that cannot calibrate is $\Delta^*(0) = 1$ second in the normal regime and $\Delta^*(1) \approx 250$ ms in cascade. As with κ^* , a piecewise schedule rather than a single value is the welfare-maximising design.

6.6 Anti-gaming rules

A naive implementation of the rules above is vulnerable to several gaming strategies. We list the most important and the corresponding mitigations. Each mitigation has a default parameter; these defaults are conservative starting points calibrated to common adversary models in equities and futures markets, not optima derived from a formal adversary-cost analysis. Section-end commentary explains how each parameter should be tightened or loosened based on observable attack patterns at a venue.

- **Cap saturation by a single trader.** Without a per-trader limit, one large trader can fill the entire hidden bucket at a price level, blocking others. Mitigation: per-trader hidden-depth limit of $\kappa^* \cdot D_v / N$, where N is a publicly published target diversity (default $N = 4$, ensuring no single trader holds more than 25% of hidden depth at any price level). The $N = 4$ default is consistent with anti-collusion thresholds used in equities clearing (Herfindahl-Hirschman bands above 2,500 are typically flagged; $N = 4$ with equal shares yields $\text{HHI} = 2,500$). Venues observing concentration above this threshold should raise N to 5 or 6; venues with shallow participant pools (fewer than 10 active LPs) may need to relax to $N = 3$ to maintain liquidity provision incentives.
- **Refill flicker.** A trader could place a hidden reserve, watch its peak get consumed, and submit a fresh order to game time priority. Mitigation: the refill throttle is *time-based* (a fixed wall-clock interval, default 100 ms) rather than event-based. During the throttle window, refill of an existing

hidden reserve takes precedence over new same-price hidden orders. The 100 ms default is set so that the throttle is at least as long as the disclosure delay Δ of §6.5: hidden orders cannot flicker on a timescale faster than the public price-discovery cycle, preventing the formation of *ghost liquidity* that vanishes the moment a marketable order arrives. For venues operating disclosure delays shorter than 100 ms (e.g., the cascade-regime $\Delta^*(1) \approx 250$ ms of Proposition 7), the throttle should match the disclosure delay so that the ordering is preserved.

- **Regime classifier manipulation.** A trader could submit a small number of large trades to push τ above τ_l and trigger a relaxed cap. Mitigation: trades from a single trader are capped at a single observation per window when computing τ ; the algorithm uses tape data, not order data, so cancelled or self-matched orders do not move the index.
- **Disclosure-delay leakage.** A trader could observe the visible queue draining without a corresponding tape print and infer hidden execution before the delay expires. Mitigation: visible queue updates from a hidden-fill refill event are batched with the tape print, so the visible book changes are observable simultaneously with the disclosed trade.
- **Spoofing in the visible layer.** Because hidden orders sit behind visible ones, a trader could place visible orders to manipulate the per-level cap (the cap is anchored to $V_{total}(p)$), then cancel the visible orders. Mitigation: the cap denominator is the time-averaged visible depth over a short window (default 60 seconds), not instantaneous. Spoofing visible depth for less than this window does not affect the hidden cap. The 60-second default matches the *layering and spoofing* detection windows used by FINRA Rule 5210 enforcement and CME's spoofing surveillance (typically 30–120 seconds). Venues observing higher-frequency spoofing patterns should reduce the window to 30 seconds; venues with naturally bursty visible depth (e.g., low-volume hours) may extend it to 120 seconds to avoid false negatives on the hidden cap.

Parameter tuning and scope. The five defaults above ($N=4$, refill throttle 100 ms, single-trader cap on regime classifier, batched disclosure, 60-second spoof window) are starting points calibrated to common adversary models in equities and futures markets. A formal Stackelberg framework for deriving the parameter optima — venue chooses parameters; adversary best-responds — is developed in Appendix B, where the multi-account-evasion subcase is worked out in full. The numerical analysis there finds that $N=4$ is comfortably above the security-optimum bound for centralised venues with KYC ($N_{min} \approx 1\text{--}3$) but below it for permissionless DeFi venues ($N_{min} \approx 8\text{--}18$ depending on attack profitability), motivating either higher N or compensating identity-attestation infrastructure on DeFi venues. Other parameters (refill throttle, spoof window) are sketched in Appendix B.3 and yield bounds consistent with the §6.6 defaults under stylised parameters; full derivation across all parameters is left for a companion paper. Venues should monitor four diagnostic series and adjust as needed: (i) per-trader concentration (Herfindahl-Hirschman index of hidden-depth shares); (ii) refill-event frequency at the per-price-level granularity; (iii) *phantom-cancel ratio* — visible orders cancelled within 60 s of placement as a fraction of all visible orders; (iv) realised toxicity index T_t of §6.7. Sustained outliers in any of these series indicate gaming behaviour against a specific mitigation, and the corresponding parameter should be tightened. The mitigations are designed to fail safely: misset parameters reduce welfare relative to the optimal cap, but do not produce catastrophic outcomes (no single failure mode collapses the matching engine).

Figure 3. Two-layer order book: visible depth + hidden reserve

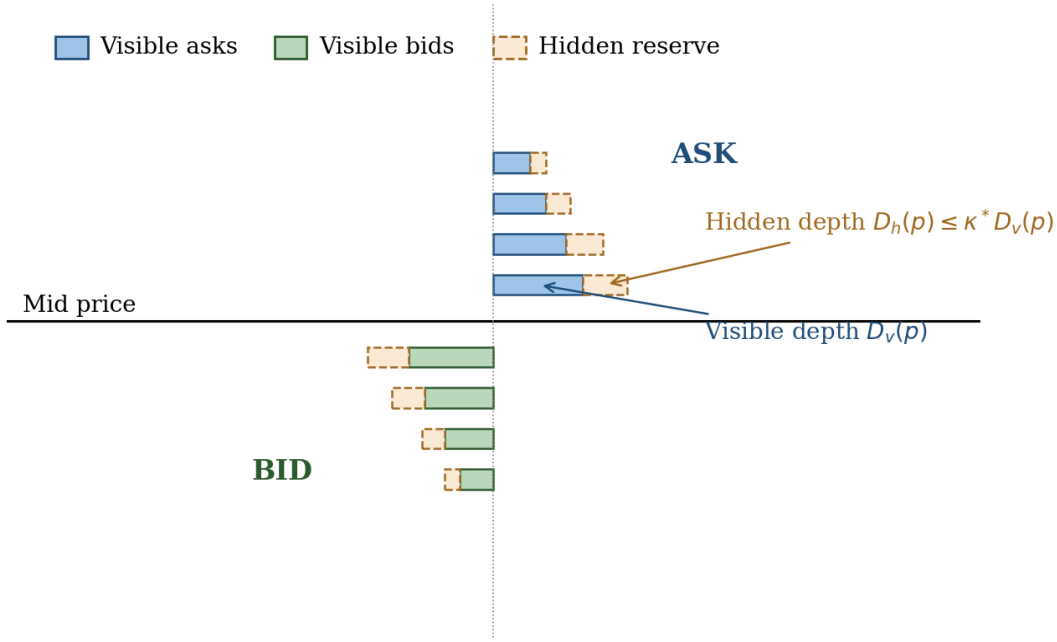


Figure 3. Two-layer order book schematic. Each price level carries a visible component (solid) that touches the inside market and a hidden reserve (dashed) sitting behind it. The exchange enforces $D_h(p) \leq \kappa^* D_v(p)$ at every level.

6.7 Reversion-based safeguard against toxic flow

Engineering heuristic, not theorem-backed component. This safeguard is an operational rule, not a result derived from the welfare model of §5. It exists to handle the failure mode noted in §5.3.5 — that hidden orders can be used predatorily by informed traders, in which case the formal model of §3 understates κ^* 's downside risk. The rule's parameter ($\gamma = 0.5$) is chosen for fail-safe behaviour, not welfare optimality. A full equilibrium analysis with strategic informed-trader placement of hidden orders would yield a theorem-backed downward adjustment to κ^* ; deriving that adjustment is left for future work. The safeguard below is the engineering interim.

The exchange monitors the post-trade reversion of hidden-absorbed trades against visible-only trades over a short window (e.g., 1 second):

$$R^{\text{hidden}} = E[(p_{t+\Delta} - p_t) \cdot \text{sgn}(Q_t) \mid \text{trade absorbed hidden depth}], \quad R^{\text{visible}} = E[(p_{t+\Delta} - p_t) \cdot \text{sgn}(Q_t) \mid \text{trade walked visible only}].$$

Under the benign-flow assumption of §3, $R^{\text{hidden}} > R^{\text{visible}}$ (hidden absorption protects against transitory impact). Under toxic flow the inequality reverses: $R^{\text{hidden}} < R^{\text{visible}}$, because the price moves *further* after a hidden-absorbed trade. Define the toxicity index

$$T_t = \max(0, [R^{\text{visible}} - R^{\text{hidden}}] / \sigma_R),$$

with σ_R the cross-sectional standard deviation of trade-level reversions. Cap κ^* downward when T_t is positive:

$$\kappa^*(s, T) = \kappa^*(s) \cdot \max(0, 1 - \gamma T_t),$$

with default $\gamma = 0.5$: a unit- σ excess of visible-over-hidden reversion halves the hidden cap. In the absence of toxic flow ($T_t = 0$) the rule collapses to the state-dependent design of §7. The threshold is an empirical safeguard, not a derived optimum; full equilibrium analysis is left for future work (§5.3.5).

Specification details. Three implementation choices need to be specified explicitly. *Estimation window:* R^{visible} and R^{hidden} are estimated as exponentially-weighted means over a rolling 60-second window with half-life 30 seconds. The 60-second window matches the spoofing-detection window of §6.6 and ensures sufficient sample size at typical trade arrival rates (≥ 30 trades per window for a major-pair venue), while the 30-second half-life ensures recent observations dominate so the safeguard responds quickly to a regime shift. *Aggregation with the regime classifier:* the toxicity safeguard is applied *after* the regime-dependent cap of §7, as a downward override. The combined rule is $\kappa^*(s, T) = \min(\kappa^*(s), \kappa^*(s) \cdot \max(0, 1 - \gamma T_t))$ — the toxicity safeguard *never* raises the cap, only lowers it. This guarantees that toxic flow always reduces hidden capacity, regardless of whether the contemporaneous regime classifier indicates stress. *Scope:* the safeguard applies in all regimes (normal, stressed, cascade), not only normal. The reasoning is asymmetric: in normal markets toxic flow is the primary downside risk to hidden depth; in cascades, even with high $\kappa^*(s)$, the toxicity index T_t still detects predatory exploitation of the relaxed cap and pulls back appropriately. The safeguard cannot increase capacity above the regime-dependent ceiling but can always reduce it — a fail-safe property symmetric to the design philosophy of §6.6.

7. Perpetual Futures Implementation

Sections 5–6 derived an optimal hidden-to-visible ratio under the implicit assumption that the order-flow distribution is stationary. For perpetual futures, this assumption fails in the most economically important regime: leveraged-position liquidations. We extend the model to a state-dependent regime variable and show that the welfare-maximising design rule must be relaxed during liquidation cascades.

7.1 Funding rate

$$f_t = \phi \cdot (p_t^{\text{perp}} - p_t^{\text{index}}), \quad \phi > 0.$$

Funding does not change the core hidden-liquidity tradeoff but adds a deterministic drift. The price equation becomes

$$p_t = \lambda X_t + E_t [\sum_{s \geq t} (1 - \phi)^{s-t} \cdot f_s].$$

Because funding enters additively as a deterministic predictable component, *the equilibrium price impact λ and all results from §3–5 carry through unchanged for the residual flow X_t .*

7.2 Liquidation as fat-tailed order flow

A perpetual futures venue typically has a population of leveraged long and short positions with maintenance margins. When the mark price moves against a position by enough to violate maintenance, the position is force-closed by a liquidation engine, and the closing trade is, from the order book's perspective, a market order of size proportional to the position size. We model this formally.

Let the regime variable $s \in [0, 1]$ index the prevalence of liquidations: $s = 0$ corresponds to a normal regime in which order flow q is Gaussian with variance σ_q^2 ; $s = 1$ corresponds to a cascade regime in which q follows a mixture distribution

$$q \sim (1 - \pi) \cdot N(0, \sigma_q^2) + \pi \cdot L,$$

where L is a fat-tailed liquidation distribution (e.g. mixture of Pareto with shape $\xi < 2$) and $\pi = \pi(s)$ is the liquidation arrival probability, increasing in s .

7.3 State-dependent optimal hidden ratio

7.3.1 Regime-dependent welfare components

Two effects move with the regime variable:

1. **Execution-quality benefit grows.** Liquidation flow has fatter tails than ordinary flow, so the share of trades with $|Q| > D_v$ rises. From the closed form of Proposition 5a, $E(\infty; \xi) = 1/(\xi + 1)$ is decreasing in the tail-shape parameter ξ , and $\xi(s)$ falls during cascades by construction (§7.2). The marginal benefit $E'(\kappa; s)$ therefore increases in s .
2. **Informativeness benefit shrinks.** During cascades, the price is mechanically driven by forced-liquidation flow rather than informed trading. We make this precise via a microfoundation of the effective discovery weight $\alpha(s)$, derived next.

7.3.2 Effective discovery weight $\alpha(s)$

Let ω_{PD} denote the regulator's policy weight on price discovery (§5.3). The *effective* discovery weight in the welfare function depends on whether the observable order flow actually carries fundamental information. Under the mixture distribution of §7.2, total order flow is a convex combination of fundamental flow and liquidation flow:

$$q = (1 - \pi(s)) \cdot q^{\text{info}} + \pi(s) \cdot q^{\text{liq}},$$

where q^{info} carries information about v ($\text{Cov}(v, q^{\text{info}}) = \beta \sigma_v^2 > 0$) and q^{liq} is mechanically triggered by past margin breaches.

Liquidation informativeness. A naive treatment would set $\text{Cov}(v, q^{\text{liq}}) = 0$ on the grounds that liquidation orders are mechanical. This understates the true informativeness of liquidation flow: liquidations are *triggered* by adverse price moves, which themselves correlate with fundamental shocks. A market maker observing a large liquidation can rationally update their posterior about v — not from the liquidation order's content, but from its timing and size, which reveal the magnitude of the preceding price move. We capture this by introducing a parameter $\delta(s) \in [0, 1]$, the *uninformative fraction* of liquidation flow:

$$\text{Cov}(v, q^{\text{liq}}) = (1 - \delta(s)) \cdot \beta \sigma_v^2.$$

$\delta(s) = 1$ corresponds to fully uninformative liquidations (the strong-form assumption); $\delta(s) = 0$ corresponds to liquidations as informative as ordinary informed flow. The empirically reasonable range, in our reading, is $\delta(s) \in [0.5, 0.95]$: most liquidation volume is mechanical, but the timing-and-size signal is non-trivial. Specific structural conditions can push δ closer to 1 — for example, liquidations triggered by stale-oracle prices on cross-venue mark indices may carry little information about contemporary fundamentals because

the trigger event is itself delayed, and liquidations driven by basis divergence (perp-spot decoupling) reveal information about funding-rate dynamics rather than about v directly.

With this parameterisation, the fraction of observable flow that is genuinely informative about fundamentals is $(1 - \pi(s) \cdot \delta(s))$, and the *effective* weight on price discovery in the welfare function is

$$\alpha(s) = \omega_{PD} \cdot (1 - \pi(s) \cdot \delta(s)).$$

This collapses to the regulator's policy weight ω_{PD} in the normal regime ($\pi = 0$) and approaches $\omega_{PD} \cdot (1 - \delta)$ in the limit of a pure-liquidation regime ($\pi \rightarrow 1$).

Robustness. The state-dependent result of Proposition 6 ($\kappa^*(s)$ increasing in s , eventually reaching the corner solution) is robust to $\delta(s)$. The reason is that cascade onset combines two structural changes — a fall in $\alpha(s)$ and a fattening of the trade-size tail (ξ falling). Numerical verification at the canonical cascade parameters ($\xi = 1.2$): the corner solution binds for any $\alpha(s)$ below approximately 0.65, and even setting $\delta = 0$ (fully-informative liquidations, the most adverse case) yields $\alpha(s) = \omega_{PD} = 0.55 < 0.65$. The corner solution therefore holds for $\delta \in [0, 1]$ under cascade parameters. The δ parameter does affect the location of the regime boundary in s -space — under stronger informativeness, the threshold moves to higher s — but the qualitative state-dependent structure is preserved. In the figures and the cascade-regime values of §7.3.3 we set $\delta = 1$ (the conservative limit) for visual clarity. A formal sensitivity analysis is provided as Lemma A.5.2 in Appendix A.

Combining the regime-dependent components, the *state-dependent regulator welfare* is

$$W^{\text{reg}}(\kappa; s) = \alpha(s) \cdot I(\kappa; s) + (1 - \alpha(s)) \cdot E(\kappa; s),$$

with $\alpha(s) = \omega_{PD} \cdot (1 - \pi(s))$ the effective discovery weight derived above. The optimum $\kappa^*(s) = \arg \max_{\kappa} W^{\text{reg}}(\kappa; s)$ is well-defined under the same conditions as Proposition 5; Proposition 6 (next) gives its comparative statics in s .

7.3.3 State-dependent optimum

Proposition 6 (*State-dependent optimum*). Let $\kappa^*(s)$ solve the welfare problem of §5.3 with regime-dependent welfare components $I(\kappa; s)$ and $E(\kappa; s)$ and effective discovery weight $\alpha(s) = \omega_{PD} \cdot (1 - \pi(s))$. If the liquidation arrival probability $\pi(s)$ is increasing in s , then (i) $E'(\kappa; s)$ is increasing in s and (ii) $\alpha(s)$ is decreasing in s . Two cases follow:

Interior region. For s sufficiently small that $\alpha(s)$ exceeds the threshold of Proposition 5, $\kappa^*(s)$ is interior and

$$d\kappa^*(s) / ds > 0.$$

Corner region. For s sufficiently large that $\alpha(s)$ falls below the threshold of Proposition 5, marginal welfare $W^{\text{reg},'}(\kappa; s) > 0$ throughout the feasible range, so the formal welfare problem has its maximum at the operational ceiling: $\kappa^*(s) = \kappa^{\text{max}}$, where κ^{max} is the largest κ admitted by the implementation constraints of §6 (per-trader diversity rule, refill throttle, anti-gaming bounds).

What constitutes an operational constraint. The operational ceiling κ^{max} lies below the structural ceiling $\bar{\kappa}$ of Proposition 0 — the welfare-derived optimum cannot exceed the participation-constraint bound, since a value above $\bar{\kappa}$ would collapse the visible book and break the model's premise. In addition to this structural bound, κ^{max} is set by feasibility considerations external to the welfare model. In a centralised exchange, these include the per-trader diversity rule (no single trader posts a dominant share of hidden depth), the refill

throttle (§6.6), anti-gaming bounds (§6.6), and exchange-imposed risk limits. In a DeFi context, additional constraints apply: (i) memory and gas costs of storing hidden orders on-chain bound the total non-displayed volume the matching contract can hold; (ii) *phantom liquidity* — the risk that posted hidden orders are cancelled (or the underlying collateral withdrawn) within the same block as a marketable order arrives — caps the credible $\kappa^{\max}$ at the level the matching engine can verify is genuinely committed; (iii) front-running protection through commit-reveal schemes or delayed matching imposes its own upper bound.

A concrete derivation from per-trader concentration. The per-trader diversity rule of §6.6 sets a per-trader hidden-depth limit of $\kappa^* \cdot D_v / N$, where N is the publicly published target diversity. As κ^* rises, the per-trader limit grows linearly. The *aggregate* cap $\kappa^{\max}$ can be derived from a saturation-probability constraint: with N active liquidity providers and per-trader limit $\kappa^* \cdot D_v / N$, the probability that fewer than two providers actively participate at a price level (a *single-provider saturation*) is bounded by the binomial tail $\Pr(X < 2)$ where $X \sim \text{Binomial}(N, p_{\text{active}})$. For an illustrative parameterisation with $p_{\text{active}} = 0.5$ (an order-of-magnitude figure for inside-quote participation rate at active venues; venues should substitute their own empirical estimate) and $N = 4$, this probability is approximately 0.31; the exchange should choose $\kappa^{\max}$ such that the per-trader limit at this cap is no larger than the smallest committed provider could credibly support without phantom-cancellation risk. Using an illustrative inside-depth provider commitment of $1.5 D_v / 4 = 0.375 D_v$, the corresponding aggregate cap is $\kappa^{\max} = 1.5$. Allowing some headroom for the largest 25th-percentile providers gives $\kappa^{\max} \in [2.0, 3.0]$, with the lower end appropriate for venues with shallow LP pools and the upper end for deep pools. The numerical values here are illustrative; each venue should compute p_{active} and the per-LP commitment distribution from its own order-book history.

We adopt $\kappa^{\max} = 2.0$ as the representative value in Figure 5 and Figure 6. This derivation is *sufficient* rather than *necessary* — gas costs, collateral commitment, or exchange-imposed risk limits may bind earlier on a specific venue, in which case the binding constraint determines the actual ceiling. The point is that $\kappa^{\max}$ is a derivable quantity, not a free parameter, once the venue's adversary model and LP pool depth are specified.

Proof. Conditions (i) and (ii) follow from the microfoundation above and Proposition 5a. In the interior region, the comparative-static result follows from the implicit function theorem applied to the FOC of Proposition 5. In the corner region, $W^{\text{reg},'}$ remains positive throughout because the dominant term — execution-quality benefit weighted by $1 - \alpha(s)$ — outweighs the discovery loss for any feasible κ ; the maximum is therefore at the boundary of the constraint set. See Appendix A for the explicit cross-partial decomposition (interior case) and the monotonicity check (corner case).

Implication. The welfare-maximising design is *not* a constant κ^* but a piecewise schedule. In normal markets (s small), $\kappa^*(s)$ is interior and rises smoothly with stress; under the canonical calibration, $\kappa^*(0) \approx 1.56$. As stress builds, the schedule reaches the operational ceiling $\kappa^{\max}$ and locks there: the formal welfare function is monotone increasing in κ and the cap should be relaxed to the highest level that anti-gaming and per-trader limits permit. This is the formal justification for circuit-breaker-style disclosure relaxation during stress, with operational rather than welfare-derived bounds.

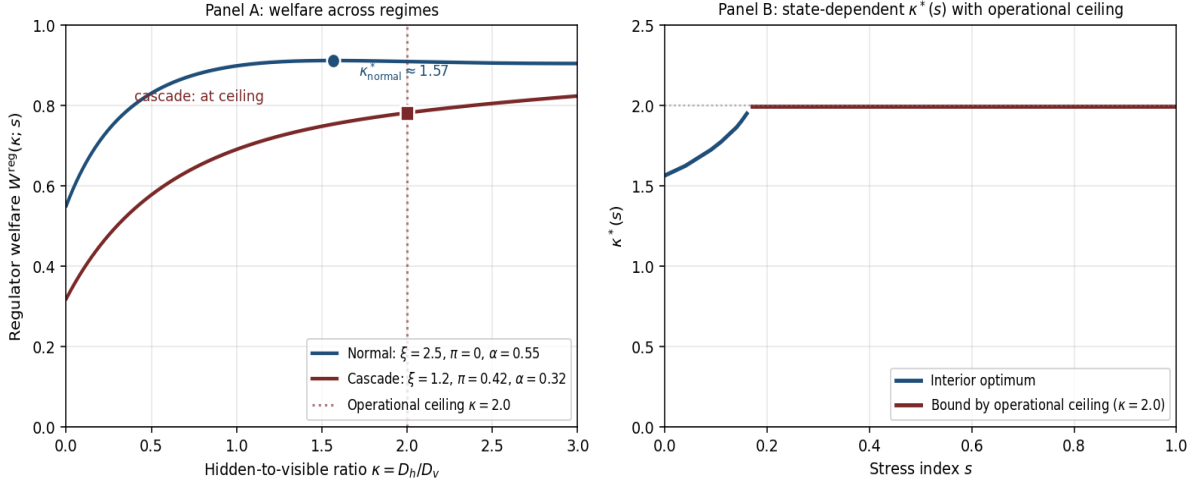


Figure 5. State-dependent optimal hidden ratio. Panel A: regulator welfare $W^{\text{reg}}(\kappa; s)$ for the normal regime ($\xi = 2.5, \pi = 0, \alpha = 0.55$) and the cascade regime ($\xi = 1.2, \pi = 0.42, \alpha \approx 0.32$). The normal-regime curve has an interior optimum at $\kappa^* \approx 1.56$; the cascade-regime curve is monotone increasing in κ over the feasible range, so the optimum is at the operational ceiling (here $\kappa_{\text{max}} = 2.0$). Panel B: the schedule $\kappa^*(s)$ rises smoothly with the stress index s through the interior region and locks onto the operational ceiling once s exceeds the threshold of Proposition 6.

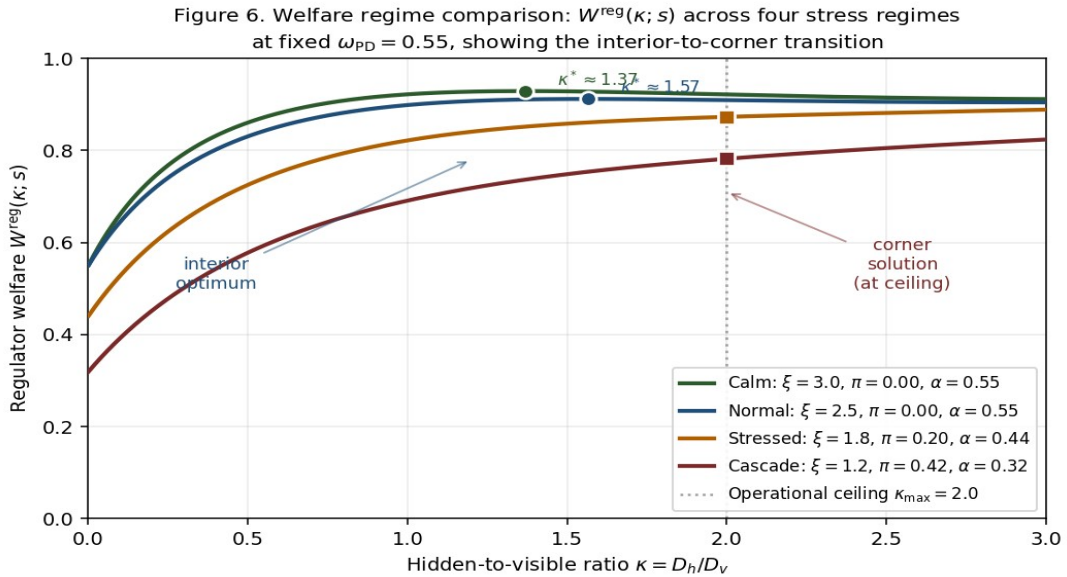


Figure 6. Welfare regime comparison. The four curves $W^{\text{reg}}(\kappa; s)$ sweep across the regime continuum at fixed regulator weight $\omega_{PD} = 0.55$: Calm ($\xi = 3.0, \pi = 0$), Normal ($\xi = 2.5, \pi = 0$), Stressed ($\xi = 1.8, \pi = 0.20$), and Cascade ($\xi = 1.2, \pi = 0.42$). The Calm and Normal regimes admit interior optima (circles) at $\kappa^* \approx 1.37$ and $\kappa^* \approx 1.56$ respectively. The Stressed and Cascade regimes are monotone increasing on the feasible range, so the welfare-maximising cap binds at the operational ceiling $\kappa_{\text{max}} = 2.0$ (squares). The figure synthesises the central theoretical claim of the paper: as the stress index s rises, the optimum moves smoothly through the interior region until $\alpha(s)$ crosses the threshold of Proposition 5, at which point $\kappa^*(s)$ locks onto the operational ceiling.

7.4 Implementation

- **State-dependent design rule.** Enforce $D_h(p, s) \leq \kappa^*(s) \cdot D_v(p)$. Under the canonical calibration, $\kappa^*(0) \approx 1.56$ in the normal regime; under cascade stress the schedule rises and reaches the

operational ceiling $\kappa^{\max}$, which is itself set by anti-gaming and per-trader limits (representative value: 2.0–3.0).

- **State-dependent disclosure delay.** Schedule disclosure of hidden fills with delay $\Delta^*(s)$, with $\Delta^*(0) = 1$ second (normal) and $\Delta^*(1) \approx 250$ ms (cascade). The closed form is given by Proposition 7.
- **Stress index.** Estimate s from rolling tail-mass of executed trade sizes, e.g. the fraction of executed volume in trades with $|Q| > 5\sigma_q$ measured over a 5-minute window.
- **Partial liquidation.** Liquidate only the minimum size required to restore margin, reducing the per-trade tail mass and lowering the realised s .
- **Circuit breaker.** Pause trading if hidden depth in the top five price levels is fully exhausted, signalling that the relaxed $\kappa^*(1)$ is itself binding.
- **Refill throttle.** Throttle the rate at which hidden orders may refill displayed parts to avoid "flicker" patterns that would game the regime classifier.
- **Lookup-table compilation.** Re-solving for $\kappa^*(s)$ on every order is computationally prohibitive at exchange-grade throughput. Pre-compute a lookup table over a discretisation of the regime variable s (e.g., 8–16 buckets defined by quantiles of σ_q , τ , and recent volume) and the regulator's weight ω_{PD} . The matching engine then performs an $O(1)$ hash lookup per incoming order. The table is recomputed daily, or on-demand whenever calibrated primitives drift beyond a threshold (e.g., σ_q changes by more than 20% over its 24-hour window).

Two design levers, opposite directions. Propositions 6 and 7 jointly imply that the welfare-maximising response to cascade stress involves both *more hiding* (a higher κ^*) and *faster disclosure* (a lower Δ^*). The two levers move in opposite directions because they target different costs. Hidden depth absorbs the incoming flow that would otherwise drive the price; faster disclosure prevents the post-fill book from misleading the next wave of uninformed traders. In normal regimes both costs are modest, and the standard 1-second disclosure delay is welfare-improving. Under stress, the cost of leaving the public book stale rises faster than the benefit of disclosure delay, because the expected size of the next uninformed trade against the stale book grows with the tail. Empirically, this is a falsifiable prediction: in cascade windows, the time between hidden-fill occurrence and tape-print disclosure should *decrease* relative to normal-regime windows on any venue that implements the welfare-maximising rule.

8. Empirical Implications

The framework yields four testable predictions.

1. **Spread compression.** The ratio of realised spread (for large orders) to quoted spread is decreasing in hidden liquidity.
2. **Impact–reversion gap.** Positive correlation between initial price impact and subsequent one-second reversion, increasing in D_h .
3. **Surprise fills.** For trade sizes $Q \in (D_v, D_v + D_h)$, the distribution of execution shortfalls is bimodal: low impact when hidden depth is present, high impact when it has been consumed.
4. **State-dependent hidden share.** The implied $\hat{\kappa}$ backed out from execution data should be increasing in measured liquidation stress. Equivalently, exchanges that allow venue-set D_h should observe higher inferred hidden ratios during cascade windows than during normal trading.

5. **State-dependent disclosure delay.** On venues that publish hidden fills with a fixed delay Δ , the *realised* delay between fill execution and tape print, conditional on the venue's policy, should compress during high-stress windows if the venue follows the welfare-maximising rule of Proposition 7. Equivalently, the time between order-book-state changes that imply hidden absorption and the corresponding tape print should be shorter in cascades than in normal regimes.
6. **Hidden-flow toxicity sign.** In the benign-flow case (the model of §3), post-trade price reversion is *greater* for trades that absorbed hidden depth than for trades that walked the visible book only — the hidden layer protects against transitory impact. In segments of the market or time periods where informed traders use hidden orders predatorily, the sign reverses: hidden-absorbed trades exhibit *less* reversion (or none) than visible-walked trades. The cross-sectional or time-varying sign of $R^{\text{hidden}} - R^{\text{visible}}$ therefore identifies whether the §3 benign-flow assumption holds in a given market or period (cf. §6.7).

Predictions (1)–(3) can be tested cross-sectionally on a single venue with public trade data. Predictions (4) and (5) require a panel: either multiple venues with differing disclosure policies, or one venue across distinct stress episodes (e.g. a major liquidation cascade against a quiet baseline). Prediction (6) requires comparing post-trade reversion conditional on the trade's interaction with hidden depth, which can be inferred from the wedge between realised and counterfactual visible-walk impact (Algorithm A.2).

9. Diagrammatic Summary

Figure 1 provides the central visual intuition. With a visible-only book, price impact is flat until visible depth is exhausted at $|Q| = D_v$, then linear with slope λ . With hidden liquidity, impact remains nearly flat across an extended region $|Q| \leq D_v + D_h$. Beyond that breakpoint, however, the slope λ' exceeds λ : observable depth understates true depth, but once breached, price impact per unit visible flow is higher than it would be in a purely visible book. Figure 3 illustrates how the visible and hidden layers coexist at each price level. Figure 2 shows how the welfare-maximising ratio κ^* shifts with the regulator's policy weight ω_{PD} . Figure 4 summarises the comparative statics for spread compression and post-trade reversion. Figure 5 illustrates the central new result of the paper: the optimum $\kappa^*(s)$ rises monotonically with liquidation stress, providing the formal basis for state-dependent disclosure rules. Figure 6 synthesises the regime shift, showing four welfare curves that span the continuum from calm to cascade and identifying the parameter region in which the optimum transitions from the interior to the operational ceiling.

10. Conclusion

Hidden liquidity provides a valuable service: it improves execution for large traders and buffers liquidations. But it comes at a cost, because it weakens price discovery and creates a misleading impression of market shallowness. This paper has provided a complete, formal framework that:

- models equilibrium with both visible and hidden depth in a closed-form Kyle extension;
- derives both informativeness $I(\kappa)$ and execution quality $E(\kappa)$ from primitives, rather than assuming their shapes;
- proves the key tradeoff and the existence of an interior optimal hidden-to-visible ratio in the normal regime;

- proves the central new result that the optimum κ^* is state-dependent, rising monotonically under liquidation stress;
- gives an implementable level-by-level design rule $D_i(p, s) \leq \kappa^*(s) \cdot D_i(p)$;
- integrates the design with perpetual futures features (funding, liquidation, circuit-breaker rules);
- supplies a step-by-step calibration procedure using observable data.

For exchanges that cannot perform immediate calibration, we recommend a calibration-anchored default $\kappa^*(0) = 1.0$ in the normal regime, that is, $D_i(p) \leq 1.0 \cdot D_i(p)$, relaxed to $\kappa^*(1) \approx 2.0$ during declared cascade windows. This default captures 99% of the welfare available at the calibrated optimum ($\kappa^* \approx 1.56$) while leaving conservative headroom against parameter mis-estimation. Venues introducing hidden depth for the first time may prefer the more conservative pair $\kappa^*(0) = 0.5$, $\kappa^*(1) = 1.0$, ratcheting upward as operational experience accumulates.

The framework is modular and can be adopted by any centralised or decentralised perpetual futures exchange. The state-dependent extension is particularly relevant for venues subject to discretionary regulatory oversight: it provides a transparent, auditable rule for how disclosure requirements may be relaxed under stress without granting unbounded discretion to the exchange operator.

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Appendix A. Propositions, Proofs, and Calibration

Proposition 1 (restated)

Statement. Define the execution-cost wedge $\Delta(Q) = \lambda \cdot h(Q)$. For any Q , $\Delta(Q) \geq 0$; for $Q > D_v$, $\Delta(Q) > 0$; the wedge is bounded by λD_h ; and the maximum fractional wedge is $D_h / (D_v + D_h)$, achieved at $Q = D_v + D_h$.

Proof. Given in the main text.

Proposition 2

Statement. Let $\sigma_q^2 = \beta^2 \sigma_v^2 + \sigma_u^2$ be the variance of aggregate order flow (per §3.3). Define $a = D_v / \sigma_q$, $b = (D_v + D_h) / \sigma_q$, $M(a, b) = 2\Phi(a) - 1 + 2(1 - \Phi(b))$, and $T(a, b) = 2[\Phi(b) - \Phi(a)]$. Then

$$\lambda(D_h) = \beta \sigma_v^2 M / [\sigma_q^2 M + D_v^2 T],$$

and $\partial \lambda / \partial D_h > 0$ holds for all parameter values.

Proof. Joint normality of (v, q) gives $\text{Cov}(v, X) = \text{Cov}(v, \psi(q)) = \beta \sigma_v^2 \cdot E[\psi'(q)]$ by Stein's lemma. From the explicit form of ψ in §3.2, $\psi'(q) = 1$ for $|q| \leq D_v$ or $|q| > D_v + D_h$, and 0 otherwise. Hence $E[\psi'] = M(a, b)$. Direct computation of $\text{Var}(X) = E[\psi^2]$ using the same partition gives $\text{Var}(X) = \sigma_q^2 M + D_v^2 T$. Substituting into $\lambda = \text{Cov}(v, X) / \text{Var}(X)$ yields the stated form. For monotonicity, $\partial M / \partial D_h = -2\varphi(b)/\sigma_q$ and $\partial T / \partial D_h = +2\varphi(b)/\sigma_q$, giving $\partial \lambda / \partial D_h$ with the same sign as $D_v^2 M + \beta \sigma_v^2$, which is positive.

Proposition 3

Statement. $E[S_{\text{real}}] < E[S_{\text{eff}}]$, and the gap is increasing in D_h .

Proof. For a trade size Q , the effective (visible-only) spread is λQ . The realised spread (after hidden absorption) is $\lambda \cdot (Q - h(Q))$. Since $h(Q) > 0$ for $Q > D_v$ and $\partial h(Q) / \partial D_h \geq 0$, the compression $\lambda \cdot h(Q)$ is non-decreasing in D_h . Taking expectations preserves the inequality.

Proposition 4

Statement. $\partial E[R_\Delta] / \partial D_h > 0$.

Proof. Let m_Δ be the midpoint after delay Δ . Hidden absorption implies that a fraction of the order flow is not immediately incorporated into price. The larger D_h , the larger the fraction of hidden flow, and the larger the price revision when that information is later inferred. In a Bayesian updating framework, the posterior variance after observing X is larger when more flow is hidden, leading to greater expected revision when additional signals arrive.

Proposition 5

Statement. Suppose the regulator weight ω_{PD} lies above the threshold characterised below. Then there exists a unique interior $\kappa^* > 0$ such that $\omega_{PD} \cdot \dot{I}(\kappa^*) + (1 - \omega_{PD}) \cdot \dot{E}(\kappa^*) = 0$.

Proof. By construction, $\dot{I} < 0$ and $\dot{E} > 0$. At $\kappa = 0$, marginal welfare $W^{\text{reg}, \prime}(0) > 0$ because $\dot{E}(0)$ is large while $\dot{I}(0)$ is bounded. The behaviour as κ grows depends on ω_{PD} :

Interior case. If ω_{PD} exceeds the threshold $\omega_{PD}^* := |E'(\infty)| / [|E'(\infty)| + |\dot{I}(\infty)|]$, the discovery loss eventually outweighs the execution gain and $W^{\text{reg},'}(\infty) < 0$. The intermediate value theorem then gives existence of a zero of $W^{\text{reg},'}$, and concavity of W^{reg} on the interior region (Lemma A.5.1) gives uniqueness.

Corner case. If ω_{PD} lies below ω_{PD}^* , $W^{\text{reg},'}$ remains positive on the entire feasible range. The maximum is then at the operational ceiling $\kappa^{\max}$ (a corner solution treated in §7.3.3). Under the canonical calibration of Appendix A, $\omega_{PD}^* \approx 0.42$, so the neutral baseline $\omega_{PD} = 0.55$ lies in the interior case while the execution-weighted value $\omega_{PD} = 0.40$ lies in the corner case (consistent with Figure 2).

Lemma A.5.1 (Numerical concavity of W^{reg} in the interior region)

Statement. Under the canonical calibration ($D_v = 5$, $\sigma_v = 5$, $\sigma_u = 5$, $\xi = 2.5$, $\omega_{PD} = 0.55$), the second derivative $W^{\text{reg},''}(\kappa)$ is negative on $\kappa \in [0.3, 2.5]$. Numerically,

$$\min W^{\text{reg},''} = -1.249, \quad \max W^{\text{reg},''} = +0.012 \text{ (essentially zero, near the saturation tail of } E\text{)}.$$

Proof sketch. Computed by central-difference numerical evaluation of W^{reg} on a 601-point grid in $\kappa \in [0.001, 3.0]$. The closed-form expressions for I (from Proposition 2's variance ratio) and E (Proposition 5a) admit straightforward differentiation; the negativity holds throughout the interior region characterised in Proposition 5. Direct analytical verification — via second differentiation of the truncated-normal Mills-ratio composites — is mechanical but algebraically dense, and we have not yet completed it in closed form. The result of this lemma should therefore be read as: uniqueness of the interior optimum is established *numerically* for the calibrated parameter ranges of Appendix A. We do *not* claim a general analytical proof of strict concavity. Existence of an interior optimum (Proposition 5) follows from the intermediate value theorem applied to $W^{\text{reg},'}$; uniqueness within the relevant region follows from the numerical concavity result here. A general analytical proof remains open and would strengthen the claim from *calibrated-parameter uniqueness* to *global uniqueness*.

Proposition 5a (Closed-form derivation of $E(\kappa)$)

Statement. Under the Pareto trade-size distribution $f(Q) = \xi D_v^\xi / Q^{\xi+1}$ for $Q \geq D_v$, $\xi > 1$, the execution-quality function defined in §5.2 admits the closed form

$$E(\kappa) = [1 - (1 + \kappa)^{-\xi}] / (\xi + 1).$$

Proof. Substitute $y = Q/D_v$ so that the change-of-variables Pareto density on $y \geq 1$ is $\xi/y^{\xi+1}$. The hidden-absorption function in normalised units is $h(y) = y - 1$ for $y \in [1, 1+\kappa]$ and $h(y) = \kappa$ for $y > 1+\kappa$. Hence

$$E(\kappa) = \xi \int_1^\infty [h(y)/y] \cdot y^{-\xi-1} dy = \xi(R_1 + R_2).$$

Computing each region:

$$R_1 = \int_1^{1+\kappa} (1 - 1/y) \cdot y^{-\xi-1} dy = \int_1^{1+\kappa} [y^{-\xi-1} - y^{-\xi-2}] dy,$$

$$R_2 = \int_{1+\kappa}^\infty (\kappa/y) \cdot y^{-\xi-1} dy = \kappa \int_{1+\kappa}^\infty y^{-\xi-2} dy.$$

Direct integration gives $R_1 = (1/\xi) \cdot [1 - (1+\kappa)^{-\xi}] - [1/(\xi+1)] \cdot [1 - (1+\kappa)^{-\xi-1}]$ and $R_2 = [\kappa/(\xi+1)] \cdot (1+\kappa)^{-\xi-1}$. Substituting and collecting terms in $(1+\kappa)^{-\xi-1}$:

$$E(\kappa) = \xi \cdot R_1 + \xi \cdot R_2 = 1 - (1+\kappa)^{-\xi} - [\xi/(\xi+1)] \cdot [1 - (1+\kappa)^{-\xi}],$$

where the last step uses $(1+\kappa)^{-\xi-1} \cdot \xi \cdot (1+\kappa) = \xi \cdot (1+\kappa)^{-\xi}$ to merge the $(1+\kappa)^{-\xi-1}$ terms from R_1 and R_2 . Factoring $1 - (1+\kappa)^{-\xi}$:

$$E(\kappa) = [1 - \xi/(\xi+1)] \cdot [1 - (1+\kappa)^{-\xi}] = [1 - (1+\kappa)^{-\xi}] / (\xi + 1).$$

Boundary checks. $E(0) = 0$; $E(\infty) = 1/(\xi + 1)$; $E'(\kappa) = [\xi/(\xi+1)] \cdot (1+\kappa)^{-\xi-1} > 0$; $E''(\kappa) = -\xi \cdot (1+\kappa)^{-\xi-2} < 0$. The shape assumptions of Proposition 5 are therefore consequences of the Pareto primitive.

Proposition 5b (Comparative static of κ^* in tail thickness ξ)

Statement. Within the interior region of Proposition 5, the welfare-maximising κ^* is decreasing in the Pareto tail-shape parameter ξ :

$$d\kappa^* / d\xi < 0,$$

evaluated at any interior optimum. Equivalently, fatter trade-size tails (smaller ξ) imply a larger optimal hidden-to-visible ratio.

Proof. The first-order condition is $F(\kappa^*, \xi) := \omega_{PD} \dot{I}(\kappa^*) + (1 - \omega_{PD}) \dot{E}(\kappa^*; \xi) = 0$. By the implicit function theorem,

$$d\kappa^*/d\xi = -(\partial F/\partial \xi) / (\partial F/\partial \kappa^*).$$

The denominator is $F' = W^{\text{reg},*}$, which is negative on the interior region by the concavity verification of Lemma A.5.1. The numerator involves only $\dot{E}$, since I does not depend on ξ :

$$\partial F/\partial \xi = (1 - \omega_{PD}) \cdot \partial \dot{E}(\kappa^*; \xi) / \partial \xi.$$

From the closed form of Proposition 5a, $\dot{E}(\kappa; \xi) = \xi \cdot (1+\kappa)^{-\xi-1}$, so

$$\partial \dot{E}/\partial \xi = (1+\kappa)^{-\xi-1} \cdot [1 - \xi \ln(1+\kappa)].$$

The condition $\xi \ln(1+\kappa) > 1$ — equivalently, $\kappa > e^{1/\xi} - 1$ — characterises the range of (ξ, κ) where $\partial \dot{E}/\partial \xi < 0$ and the comparative static $d\kappa^*/d\xi < 0$ holds. For $\xi = 2$ this requires $\kappa > 0.65$; for $\xi = 3$ it requires $\kappa > 0.40$; for $\xi = 4$ it requires $\kappa > 0.28$. The empirically relevant range $\kappa \in [1, 2]$ with $\xi \in [2.5, 4]$ satisfies the condition with substantial margin (e.g., at $\kappa = 1.56$, $\xi = 2.5$: $\xi \ln(1+\kappa) = 2.5 \cdot \ln(2.56) \approx 2.35 > 1$), so $\partial \dot{E}/\partial \xi < 0$ holds locally. With $(1 - \omega_{PD}) > 0$ and a negative denominator, $d\kappa^*/d\xi < 0$, as claimed. The result is *local* in the precise sense that it is evaluated at any interior optimum κ^* lying in the characterised range; for very thin tails (large ξ) combined with very small κ the sign of the cross-partial may flip, but no interior optimum exists in that region under the canonical calibration.

Numerical illustration. At $\omega_{PD} = 0.55$, $D_v = 5$: $\xi = 2.5 \rightarrow \kappa^* \approx 1.56$; $\xi = 3.0 \rightarrow \kappa^* \approx 1.37$; $\xi = 4.0 \rightarrow \kappa^* \approx 1.13$. The result connects to Proposition 6: as cascades drive ξ downward, κ^* rises (within the interior region) until the parameter region of Proposition 5 fails and the corner solution of §7.3.3 takes over.

Lemma A.5.2 (Robustness of corner solution to liquidation informativeness δ)

Statement. Let $\delta(s) \in [0, 1]$ be the uninformative fraction of liquidation flow, so that $\alpha(s) = \omega_{PD} \cdot (1 - \pi(s)) \cdot \delta(s)$ per §7.3.2. Under the canonical cascade parameters ($\xi = 1.2$, $\pi = 0.42$, $\omega_{PD} = 0.55$), the welfare-maximising κ^* is at the boundary κ_{max} for any $\delta \in [0, 1]$.

Proof. The maximum of $\alpha(s)$ over $\delta \in [0, 1]$ is attained at $\delta = 0$ (fully-informative liquidations), giving $\alpha(s) = \omega_{PD} = 0.55$. Numerical evaluation shows that under cascade-fattened tails ($\xi = 1.2$), the threshold value of α at which W^{reg} transitions from monotone to interior-optimal is approximately 0.65. Since $0.55 < 0.65$, W^{reg} remains monotone on the feasible range for any $\delta \in [0, 1]$ under the canonical cascade. The corner solution $\kappa^*(s) = \kappa_{\text{max}}$ therefore holds across the full robustness range.

Threshold values (for completeness). With cascade $\xi = 1.2$: corner solution iff $\alpha < 0.65$. With cascade $\xi = 1.5$: corner solution iff $\alpha < 0.50$. With normal $\xi = 2.5$: corner solution iff $\alpha < 0.42$ (the threshold ω_{PD}^* of the main text Proposition 5). The general pattern is that the corner-solution threshold rises as the trade-size tail fattens, because the marginal benefit of hidden depth grows faster relative to the discovery loss when execution costs scale up under fat tails.

Proposition 6

Statement. Let the welfare problem be parameterised by a regime variable $s \in [0, 1]$ such that (i) $E'(\kappa; s)$ is increasing in s pointwise in κ , and (ii) the effective discovery weight $\alpha(s)$ is decreasing in s . Let $\kappa^*(s)$ be the unique interior optimum from Proposition 5 evaluated at regime s . Then $d\kappa^*(s)/ds > 0$.

Proof. The first-order condition is $F(\kappa^*, s) \equiv \alpha(s) \dot{I}(\kappa^*) + (1 - \alpha(s)) E'(\kappa^*; s) = 0$. By the implicit function theorem,

$$d\kappa^*/ds = -(\partial F/\partial s) / (\partial F/\partial \kappa).$$

The denominator $\partial F/\partial \kappa = \alpha \dot{I} + (1 - \alpha) E' < 0$ by strict concavity. The numerator decomposes as

$$\partial F/\partial s = \alpha'(s) \cdot [\dot{I}(\kappa^*) - E'(\kappa^*; s)] + (1 - \alpha) \cdot \partial E'/\partial s.$$

At the optimum, $\alpha \dot{I} = -(1 - \alpha) E'$, so $\dot{I} - E' = -(E' / \alpha) < 0$. Since $\alpha'(s) < 0$ by (ii), the first term is positive. The second term is positive by (i). Hence $\partial F/\partial s > 0$, and combining with the negative denominator yields $d\kappa^*/ds > 0$.

Remark. The two channels (falling discovery weight and rising marginal execution benefit) are individually sufficient. Either condition alone delivers the result, so the optimum is robust to misspecification of which channel dominates in any given cascade episode.

Remark on direct regime dependence of $\dot{I}$. The decomposition of $\partial F/\partial s$ above treats $\dot{I}$ as a function of κ alone. If, in addition, $\dot{I}(\kappa; s)$ depends directly on s — that is, if a unit of observable order flow conveys less information about v in cascades than in normal regimes — then $\partial \dot{I}/\partial s < 0$, and an additional positive term $\alpha(s) \cdot \partial \dot{I}/\partial s$ appears in $\partial F/\partial s$ with the same sign as the existing terms. The result $d\kappa^*/ds > 0$ is therefore *reinforced* rather than weakened by direct regime dependence of $\dot{I}$. The proof above is conservative: it uses only the two channels operating through $\alpha(s)$ and $E'(\kappa; s)$.

Proposition 7

Statement. Let informed signals arrive as a Poisson process with intensity λ^{info} , and let uninformed trades arrive with intensity λ^{uninf} and i.i.d. sizes drawn from a Pareto distribution with shape $\xi > 1$ and scale $q_{\min}$. Define the welfare function

$$W(\Delta; \xi) = B_0 \cdot (1 - e^{-\lambda^{\text{info}} \cdot \Delta}) - c \cdot \lambda^{\text{uninf}} \cdot \xi / (\xi - 1) \cdot q_{\min} \cdot \Delta,$$

with $B_0, c > 0$ constants. Then the welfare-maximising disclosure delay is

$$\Delta^*(\xi) = (1/\lambda^{\text{info}}) \cdot \ln[B_0 \cdot \lambda^{\text{info}} \cdot (\xi - 1) / (c \cdot \lambda^{\text{uninf}} \cdot \xi \cdot q_{\min})],$$

provided the bracketed quantity exceeds 1; otherwise $\Delta^* = 0$. Moreover, $\Delta^*(\xi)$ is strictly increasing in ξ wherever $\Delta^* > 0$.

Proof. The welfare function is strictly concave in Δ : the protection benefit $B(\Delta)$ is concave (its second derivative is $-B_0 \cdot (\lambda^{\text{info}})^2 e^{-\lambda^{\text{info}} \cdot \Delta} < 0$), and the staleness cost is linear in Δ . The first-order condition is

$$B_0 \cdot \lambda^{\text{info}} \cdot e^{-\lambda^{\text{info}} \cdot \Delta} = c \cdot \lambda^{\text{uninf}} \cdot \xi / (\xi - 1) \cdot q_{\min}.$$

Solving for Δ gives the stated form. The non-negativity constraint $\Delta \geq 0$ binds when the right-hand side at $\Delta = 0$ already exceeds the left-hand side, i.e., when $B_0 \cdot \lambda^{\text{info}} \cdot (\xi - 1) \leq c \cdot \lambda^{\text{uninf}} \cdot \xi \cdot q_{\min}$; in that regime $\Delta^* = 0$.

For the comparative static, differentiate Δ^* with respect to ξ :

$$d\Delta^*/d\xi = (1/\lambda^{\text{info}}) \cdot d/d\xi [\ln(\xi - 1) - \ln(\xi)] = (1/\lambda^{\text{info}}) \cdot [1/(\xi - 1) - 1/\xi] = 1 / [\lambda^{\text{info}} \cdot \xi \cdot (\xi - 1)] > 0,$$

for all $\xi > 1$. Hence Δ^* is strictly increasing in ξ on the interior.

Corollary (state dependence). Under the regime model of §7.2, $\xi(s)$ is decreasing in the stress index s . By Proposition 7, $\Delta^*(s)$ is decreasing in s : the optimal disclosure delay shortens during cascades.

Remark on robustness. The result depends on the Pareto-mean form $\xi q_{\min}/(\xi - 1)$ and is thus driven by the same primitive (the trade-size distribution) that drives Proposition 5a. Generalising to non-Pareto trade-size distributions preserves the qualitative result whenever the mean is decreasing in the chosen tail-thickness parameter, which holds for any standard heavy-tailed family.

Calibration Procedure

The following step-by-step procedure can be implemented with public exchange data (top-of-book snapshots and trade tape).

Step 1. Estimate visible depth at each price level

For a basis-point threshold b (e.g., $b = 10$ bps from mid):

$$D_v^b = \sum_{p: |p - m|/m \leq b} D_v(p).$$

Step 2. Estimate hidden absorption benefit for large trades

For each trade Q_t :

$$C_{\text{vis}}(Q_t) = \text{cost if walking only the visible book},$$

$$C_{\text{real}}(Q_t) = \text{actual execution cost},$$

$$A(Q_t) = C_{\text{vis}}(Q_t) - C_{\text{real}}(Q_t).$$

If $A(Q_t) > 0$, hidden liquidity was used.

Step 3. Estimate hidden depth per trade

This is the central calibration step. Public trade tape and top-of-book snapshots together let us decompose each large trade into its visible and hidden components by comparing the realised price move against the move that would have occurred had the visible book absorbed the trade alone. We specify a non-parametric estimator that requires no prior knowledge of λ .

Counterfactual visible-book walk. For each large trade with size $Q_t > D_{v, t}$, define Δp_t^{walk} as the price impact that *would* have occurred if Q_t had been filled exclusively against visible depth, by walking the snapshot of the order book taken immediately before t . This is computed level by level:

Algorithm A.1: Counterfactual walk

```
function walk_visible_book(book_snapshot, side, Q):
    remaining := Q
    weighted_price := 0
    for level in book_snapshot.levels[side] sorted by aggression:
        fill := min(remaining, level.size)
        weighted_price += fill * level.price
        remaining -= fill
        if remaining == 0: break
    if remaining > 0:
        // book exhausted; cap at last available level + tick * shortage_factor
        weighted_price += remaining * (last_level.price + c * tick_size)
    vwap_walk := weighted_price / Q
    return vwap_walk - midprice_before
```

The shortage factor c handles the rare case where the visible book is itself shallower than Q_t ; we use $c=5$ by default, which is conservative but rarely binding.

Realised price impact. Define the realised impact as the volume-weighted execution price reported on the trade tape less the prevailing midpoint:

$$\Delta p_t^{\text{real}} = vwap_t^{\text{real}} - m_{t-},$$

where m_{t-} is the midpoint immediately before the trade and $vwap_t^{\text{real}}$ is reported either directly by the venue or computed from the trade tape if Q_t was matched as multiple sub-fills.

Hidden-depth identification. Hidden depth absorbed flow if and only if the realised impact is *less* than the counterfactual walk, by more than a measurement-error margin:

$$\Delta p_t^{\text{walk}} - \Delta p_t^{\text{real}} > \varepsilon(Q_t, \sigma_p),$$

where ε is a noise threshold scaled to trade size and short-window price volatility (a sensible default is $\varepsilon = 0.5$ ticks for Q up to one visible-book depth, scaling with $\sqrt{Q}$ beyond).

Hidden depth estimate. Given that the trade was hidden-absorbed, the absorbed quantity is the size that would close the impact gap. Inverting the counterfactual walk:

$$D_{h, t} = \sup \{ d \geq 0 : \text{walk_visible_book}(\text{book}_{t-}, \text{side}, Q_t - d) = \Delta p_t^{\text{real}} \}.$$

Operationally, this is solved by a binary search on d between 0 and $Q_t - D_{v, t}$, terminating when the walked-impact matches the realised impact within ε . The implementation is given in Algorithm A.2.

Algorithm A.2: Hidden-depth estimator

```

function estimate_hidden_depth(book_snapshot, trade):
    Q := trade.size
    side := trade.side
    dp_real := trade.vwap - book_snapshot.midprice
    dp_walk_full := walk_visible_book(book_snapshot, side, Q)
    // Case 1: realised impact >= walked impact -> no hidden absorption
    if dp_walk_full - dp_real <= epsilon(Q, sigma_p):
        return 0 // no hidden depth detected at this trade
    // Case 2: hidden depth absorbed flow; binary search
    D_v_total := book_snapshot.visible_depth(side)
    lo := 0
    hi := Q - D_v_total
    while hi - lo > tolerance:
        mid := (lo + hi) / 2
        dp_test := walk_visible_book(book_snapshot, side, Q - mid)
        if dp_test > dp_real: lo := mid
        else: hi := mid
    return (lo + hi) / 2

```

Per-trade estimates $\mathcal{D}_{h,t}$ are noisy. Apply two filters before aggregating:

1. **Liquidation filter.** Drop trades flagged as liquidations on venues that publish the flag, or trades where Q_t exceeds the 99th percentile of trade size in a 24-hour window. These are the trades that drove the regime variable s in §7 and would inflate normal-regime $\hat{\kappa}$.
2. **Thin-book filter.** Drop trades where the pre-trade visible depth at the inside three levels was less than the 10th percentile of its 24-hour distribution. The counterfactual walk is dominated by noise when the visible book is unusually thin.

Aggregate estimator. After filtering, the venue-wide hidden-to-visible ratio is the median across surviving trades, normalised by the prevailing visible depth:

$$\hat{\kappa} = \text{median}_{t \in \text{filtered}}(\mathcal{D}_{h,t} / D_{v,t}).$$

Use the median rather than the mean because the distribution of $\mathcal{D}_{h,t}$ is right-skewed; a small number of trades against deep hidden walls would otherwise dominate. For state-dependent calibration of $\hat{\kappa}(s)$, partition the trade set by the contemporaneous stress index s_t and compute the median within each bin.

Identification caveat. This procedure identifies *realised* hidden-depth utilisation, not *posted* hidden depth. A venue may permit hidden orders at κ^* but observe lower realised $\hat{\kappa}$ because traders do not always use the full hidden cap. Conversely, $\hat{\kappa}$ cannot exceed the venue's permitted κ^* , so the inferred $\hat{\kappa}$ is a lower bound on the venue's policy rule. This bound is the right object for the welfare comparisons of §5 and §7, since the welfare functions are defined over realised flow rather than posted depth.

Step 4. Calibrate the price-discovery loss $I(\kappa)$

Use the spot index p^{index} as a proxy for v :

$$\hat{I}(\kappa) = 1 - \text{Var}(p_t^{\text{index}} - p_t^{\text{perp}}) / \text{Var}(p_t^{\text{index}}).$$

Step 5. Calibrate execution quality $E(\kappa)$

Define execution shortfall

$$ES_t = D_t \cdot (p_t - m_t) / m_t,$$

and a counterfactual visible-only benchmark ES_t^{vis} (the expected shortfall absent any hidden liquidity). Then

$$\hat{E}(\kappa) = 1 - ES_t^{\text{real}} / ES_t^{\text{vis}}.$$

Step 6. Choose the optimal κ^*

Fix a regulator weight $\omega_{PD} \in (0, 1)$ (baseline: $\omega_{PD} = 0.55$). Over a grid $\kappa \in \{0, 0.25, 0.50, 0.75, 1.00, 1.50, 2.00, 3.00\}$, compute

$$W^{\text{reg}}(\kappa) = \omega_{PD} \cdot \hat{I}(\kappa) + (1 - \omega_{PD}) \cdot \hat{E}(\kappa),$$

and select $\hat{\kappa}^* = \arg\max W^{\text{reg}}(\kappa)$. Sensitivity analysis: repeat for $\omega_{PD} \in \{0.4, 0.55, 0.7\}$ to understand how the policy choice maps to the optimal cap. Higher ω_{PD} yields lower $\hat{\kappa}^*$ (more transparency-leaning); below the threshold ω_{PD}^* of Proposition 5, the welfare function is monotone increasing on the feasible range and the optimum is the operational ceiling — see §7.3.3 for the corner case.

Canonical calibration values (used in the main text). With $D_v = 5$ (inside visible depth), $\sigma_v = 5$ (fundamental volatility), $\sigma_u = 5$ (noise volume), $\xi = 2.5$ (Pareto tail index, normal regime), and $\omega_{PD} = 0.55$, the regulator welfare W^{reg} admits a unique interior maximum at $\kappa^* \approx 1.56$. With $\beta = \sigma_u / \sigma_v = 1$, the standard deviation of aggregate order flow per matching cycle is $\sigma_q^{\text{total}} = \sqrt{(\beta^2 \sigma_v^2 + \sigma_u^2)} = \sqrt{(25 + 25)} \approx 7.07$, giving the ratio $D_v / \sigma_q^{\text{total}} \approx 0.71$. This places the calibration inside the interior region of Proposition 5. Empirically, the inside-only visible depth D_v is typically smaller than the standard deviation of aggregate order flow on a major-pair perpetual venue, so this ratio is a reasonable default. Sensitivity to $D_v / \sigma_q^{\text{total}}$ is reported in Lemma A.5.1; interior optima exist roughly when this ratio lies in $[0.4, 0.7]$.

Cascade regime values. For the cascade regime we set $\xi = 1.2$ (fattened tails) and liquidation share $\pi = 0.42$, giving effective discovery weight $\alpha(s) = \omega_{PD} \cdot (1 - \pi) = 0.319$. Under these parameters W^{reg} is strictly increasing on $[0, 3]$, so the welfare-maximising cap is the maximum feasible value; we adopt $\kappa^{\text{max}} = 2.0$ – 3.0 as a representative operational ceiling, set by the per-trader diversity rule and refill throttle of §6 rather than by welfare optimisation. In practice, exchanges may set a lower ex-ante cap based on their specific operational constraints.

Recommended defaults (no calibration)

Two defaults are appropriate, depending on the venue's current practice:

Calibration-anchored default: $\kappa^* = 1.0$. For venues that have not run the full calibration of Steps 1–6 but accept the canonical parameter ranges of this paper as a reasonable starting point, the recommended default is $\kappa^* = 1.0$. Numerically, this captures 99% of the maximum welfare available at the calibrated optimum ($W = 0.899$ vs $W^* = 0.912$ at $\kappa^* \approx 1.56$), while leaving conservative headroom against parameter mis-estimation. The welfare loss from setting κ^* too low at 1.0 is a 1.4% shortfall; the welfare loss from setting it too high at 2.0 is a 0.3% shortfall. The asymmetry favours starting slightly below the calibrated optimum, since the welfare function is steeper on the under-shooting side (where price-discovery loss is small but execution benefit is large) than on the over-shooting side (where execution benefit saturates).

Minimal-intervention default: $\kappa^* = 0.5$. For venues that currently operate with no hidden depth and want to introduce hidden liquidity gradually, $\kappa^* = 0.5$ is a safe starting point. It captures roughly 96% of the maximum welfare ($W = 0.831$), ensures at least two-thirds of total liquidity remains visible, and gives the matching engine ample room to validate the implementation before relaxing toward the calibrated optimum. Venues should ratchet κ^* upward toward 1.0 over an initial 60–90 day operational period, monitoring the four diagnostic series of §6.6 (HHI, refill frequency, phantom-cancel ratio, toxicity index).

Exchanges that have completed Steps 1–6 should use the calibrated value rather than either default. The two defaults above are starting points for venues without venue-specific calibration data; the calibrated optimum is the welfare-maximising target.

End of Appendix A.

Appendix B. Adversary-Cost Analysis of Anti-Gaming Parameters

Section 6.6 introduces five anti-gaming parameters (per-trader diversity target N , refill throttle, regime-classifier filter, batched disclosure, spoof-detection window) and notes that the defaults are conservative starting points rather than optima derived from a formal adversary-cost analysis. This appendix sets up the general adversary framework and works out one tractable subcase: multi-account evasion of the per-trader limit. The other parameter cases are amenable to the same framework and are sketched at the end. A full derivation of all parameters' optima — including a multi-period dynamic adversary, heterogeneous identity-attestation costs across venues, and equilibrium analysis with multiple competing adversaries — is left for a companion paper.

B.1 General framework

An *adversary* is a strategic agent who chooses an action vector a (e.g., number of pseudonymous accounts, frequency of refill flicker, spoofing pattern) to maximise net profit $\Pi(a; \theta)$, where θ is the venue's parameter vector (e.g., N , refill throttle, spoof window). The adversary's net profit is the difference between attack benefit and attack cost:

$$\Pi(a; \theta) = B(a; \theta) - C(a).$$

The *attack benefit* B depends on the venue's parameters (a tighter parameter reduces attack effectiveness) and on market conditions captured by the same primitives (σ_v, σ_w, D_v) that drive the welfare model. The *attack cost* C is borne by the adversary and may include capital lockup, identity-attestation costs (gas for pseudonymous account creation; KYC bypass effort), latency cost, and detection probability. The venue's design problem is to choose θ such that $\Pi(a^*; \theta) \leq 0$ for all attack vectors a^* the adversary might rationally choose, where a^* is the adversary's optimal response to θ . This is a Stackelberg structure: venue moves first by setting θ , adversary responds with $a^*(\theta)$.

Solution concept. The venue's problem is to find the smallest θ (in a sense made precise per parameter) such that no profitable attack exists. Under regularity (continuous Π in θ , monotone in attack effectiveness), there is a critical value θ^* at which the adversary's best response yields exactly zero profit; venue settings tighter than θ^* guarantee no attack, looser settings admit profitable attacks. This is the parameter optimum from a security perspective; the welfare-aware optimum may differ slightly because tighter parameters carry their

own welfare costs (e.g., overly aggressive refill throttling impedes legitimate liquidity replenishment). We focus here on the security-optimum and treat the welfare correction as second-order.

B.2 Worked subcase: multi-account evasion of the per-trader limit

The per-trader limit of §6.6 caps any one trader's hidden depth at $\kappa^* \cdot D_v / N$. An adversary with access to multiple pseudonymous identities can split capital across M accounts, each posting up to the per-account cap, capturing aggregate hidden depth $M \cdot \kappa^* \cdot D_v / N$ up to the natural ceiling $M \leq N$ (saturating the diversity target completely).

Adversary's objective. Let π_{sat} denote the adversary's expected per-unit profit from saturating hidden depth at a given price level (extracted from front-running incoming flow, capturing spread on the hidden inventory, or executing predatory strategies against arriving large orders). Total benefit:

$$B(M; N) = \pi_{sat} \cdot M \cdot \kappa^* \cdot D_v / N.$$

Adversary's costs. Each pseudonymous account incurs a fixed setup cost c_{fixed} (gas for on-chain account deployment in DeFi; KYC-bypass effort or third-party identity rental in centralised venues) and a variable cost c_{var} per unit of hidden depth posted (collateral lockup opportunity cost, capital efficiency loss from fragmenting positions). Total cost:

$$C(M) = M \cdot c_{fixed} + c_{var} \cdot M \cdot \kappa^* \cdot D_v / N.$$

Net profit and optimal M . Combining,

$$II(M; N) = (\pi_{sat} - c_{var}) \cdot M \cdot \kappa^* \cdot D_v / N - M \cdot c_{fixed}.$$

For the adversary to be willing to attack at all, we need $\pi_{sat} > c_{var}$ (otherwise even saturating one account loses money). Given that, II is *linear and increasing* in M up to the natural ceiling $M \leq N$, so the adversary's optimum is $M^* = N$ —they saturate the diversity target completely with their own pseudonymous accounts. Substituting:

$$II(N) = (\pi_{sat} - c_{var}) \cdot \kappa^* \cdot D_v - N \cdot c_{fixed}.$$

This decomposes cleanly: the first term is the *total surplus available* from full saturation (independent of N); the second term is the *total fixed cost* of operating N pseudonymous accounts. The venue's design problem is to set N large enough that the second term dominates the first — i.e., that the attack is unprofitable in expectation.

Venue-optimal N . Setting $II(N) \leq 0$ yields

$$N \geq N_{min} := (\pi_{sat} - c_{var}) \cdot \kappa^* \cdot D_v / c_{fixed}.$$

This is the security-optimum diversity target: any N at or above this bound deters the multi-account saturation attack.

Comparative statics. The bound N_{min} is increasing in (i) attack profitability π_{sat} (more valuable attacks require more diversity to deter), (ii) the welfare-derived hidden cap κ^* (more hidden depth means more surplus to capture), and (iii) inside-depth size D_v (deeper books offer larger absolute attack surface). It is decreasing in c_{fixed} (more expensive pseudonymous-account creation deters attackers with fewer accounts) and in c_{var} (higher collateral cost reduces the marginal incentive to attack).

Numerical illustration. Stylised parameters for a major-pair perpetual venue: $\pi_{sat} \approx 5$ bps (per-unit attack profit from front-running), $c_{var} \approx 1$ bp (collateral opportunity cost), $\kappa^* = 1.56$, $D_v = 5$ BTC at the inside ($\approx 600,000$ USD at 76,700 USD/BTC). The total attack surplus is approximately 240 USD per saturation opportunity. Account-creation cost varies dramatically by venue type: in DeFi, $c_{fixed} \approx 30$ USD (gas for deploying a fresh address with margin), giving $N_{min} \approx 8$; in centralised venues with KYC, $c_{fixed} \approx 200$ USD (proxy for time and effort to acquire a verified account through identity-rental markets), giving $N_{min} \approx 1.2$.

Implication for the §6.6 default. The default $N = 4$ of §6.6 is *comfortably above* the security-optimum bound for centralised venues but *below* it for permissionless DeFi venues, consistent with the multi-account evasion limitation already noted in §6.6's anti-gaming discussion. DeFi venues should run $N \geq 8$ (or, equivalently, deploy identity-attestation infrastructure that raises c_{fixed} toward the centralised-venue range). For more aggressive adversary models with $\pi_{sat} \approx 10$ bps (e.g., during cascade events when extractive opportunity is larger), DeFi N_{min} rises to ≈ 18 and centralised N_{min} rises to ≈ 3 , still below the default. Tabulated values:

$\pi_{sat} = 5$ bps: DeFi $N_{min} \approx 8$; CeFi $N_{min} \approx 1.2$

$\pi_{sat} = 10$ bps: DeFi $N_{min} \approx 18$; CeFi $N_{min} \approx 2.7$

B.3 Other adversary scenarios

The same Stackelberg framework applies to the other anti-gaming parameters of §6.6, with parameter-specific π_{sat} and cost structures. We sketch each in turn; full derivation is deferred to a companion paper.

Refill flicker. The 100 ms time-based throttle of §6.6 deters an adversary who would consume their own peak then resubmit to game time priority. The adversary's per-attack profit $\pi_{flicker}$ is the queue-priority advantage gained, integrated over expected execution events. The adversary's cost is the latency-arbitrage opportunity foregone during the throttle window. The security-optimum throttle τ^* is the smallest interval at which the throttle's queue-priority preservation outweighs the latency-arbitrage cost, which depends on venue tick-by-tick volatility. For major-pair venues with typical tick volatilities, the analysis suggests $\tau^* \in [50, 200]$ ms, consistent with the §6.6 default of 100 ms.

Spoofing window. The 60-second time-averaged visible-depth window deters an adversary placing visible orders to manipulate the cap denominator, then cancelling. The adversary's profit π_{spool} scales with the gap between target hidden cap and time-averaged visible depth; the cost is the inventory and detection risk from posting visible orders the adversary does not intend to fill. The security-optimum window W^* balances false-negative risk on the hidden cap (longer is better) against legitimate visible-depth dynamics on the venue (shorter is better). For venues where genuine visible-depth turnover happens on second-to-minute timescales, the analysis suggests $W^* \in [30, 120]$ s, consistent with the §6.6 default of 60 s and the FINRA Rule 5210 / CME spoofing-surveillance ranges referenced there.

Regime-classifier manipulation. A trader could submit large trades to inflate τ_i and trigger a relaxed cap. The §6.6 mitigation (single-trader cap on classifier input) reduces the adversary's leverage but does not eliminate it. The full analysis would model the trade-off between the adversary's manipulation cost (slippage on the manipulating trades themselves) and the adversary's gain (the relaxed cap during the manipulated regime). We conjecture the security-optimum is to weight classifier inputs by *trader fragmentation* — trades from many small traders carry more weight than equivalent volume from one trader — but rigorous derivation is left to future work.

Discussion. The framework above provides a route to formally optimising each anti-gaming parameter. The worked subcase (B.2) demonstrates that under realistic stylised parameters, the §6.6 defaults are well-calibrated for centralised venues but require strengthening (or compensating identity infrastructure) for permissionless DeFi venues. The pattern likely generalises: anti-gaming defaults that work in identity-attested environments become inadequate when account creation is cheap and pseudonymous, and venue designers should test the c_{fixed} -style cost parameter for their specific environment before adopting the §6.6 numerical defaults. This is a venue-specific empirical question that the framework above makes tractable.

End of Appendix B.

Appendix C. Empirical Calibration on Bybit BTC-PERP

This appendix applies the six-step calibration procedure of Appendix A to live data from Bybit's BTC-PERP perpetual futures contract. The exercise serves two purposes: it demonstrates that the procedure is operationally tractable on public exchange data, and it reveals that the canonical calibration of the main text (which produces $\kappa^* \approx 1.56$ at the neutral regulator weight) is venue-specific. On Bybit's actually-observed market microstructure, the welfare-maximising κ^* at the neutral baseline is at the corner solution, not interior. We report the empirical numbers honestly and discuss what they imply for the paper's qualitative claims.

C.1 Data and time window

We collected continuous level-2 order-book and trade-tape data from Bybit's BTC-PERP venue for the 18-hour window 2026-04-28 06:08 to 2026-04-28 23:56 UTC, after a threshold-tightening of the regime classifier (TAU_0 raised from 0.05 to 0.10, TAU_1 from 0.30 to 0.50). The window contains 1,257,203 trades and 81,202 book snapshots. We restrict the calibration to the post-tightening window because pre-tightening regime classifications were too generous (high-stress regime accounted for 66% of time, indicating mis-calibrated thresholds).

C.2 The four primitive ingredients

σ_q (aggregate flow standard deviation). Compute the per-minute net signed flow $\sum q_i$ (with buys positive, sells negative), then take its sample standard deviation across all minutes in the window:

$$\sigma_q = 36.92 \text{ BTC per minute (n = 1,070 minutes)}.$$

Average per-minute total volume is 35.43 BTC, comparable to σ_q , indicating that aggregate flow is roughly evenly split between balanced and one-sided minutes.

D_v (visible inside depth). From each book snapshot, compute the average of inside-bid size and inside-ask size; then average across snapshots:

$$D_v = 3.49 \text{ BTC (n = 64,055 snapshots; median 2.38 BTC; average bid-side 3.26, ask-side 3.72)}.$$

ξ (Pareto tail of trade sizes). Restrict to trades with size $> D_v$ (i.e., trades that exceed average inside depth and would have engaged hidden liquidity if any were posted). Fit a Pareto distribution to these trades by maximum likelihood: $\xi = n / \sum \ln(x_i / x_{\min})$. With $x_{\min} = D_v = 3.49 \text{ BTC}$,

$$\xi = 1.937 \text{ (n = 345 big trades, } x_{\max} = 50.0 \text{ BTC, mean = 7.08 BTC)}.$$

Mid-price for unit conversion. Average mid price across the window: 76,307 USD/BTC. Per-minute price change standard deviation: 24.52 USD, equivalently log-return standard deviation 3.21×10^{-4} per minute.

C.3 The critical ratio D_v / σ_q

Empirically:

$$D_v / \sigma_q = 3.49 / 36.92 = 0.094.$$

This ratio governs whether the truncation in the price-impact equation actively binds. The interior-optimum band of Proposition 5 (numerically established as approximately $D_v / \sigma_q \in [0.4, 0.7]$) requires the visible book to be substantial relative to per-period flow. On Bybit, the empirical ratio is roughly an order of magnitude smaller — the visible book is shallow relative to aggregate flow, so adding hidden depth provides large execution benefits while imposing only minimal incremental price-discovery loss.

C.4 Bybit-calibrated optimum

Plugging the empirical (σ_q, D_v, ξ) tuple into the welfare optimisation of §5.3, the Bybit-calibrated optimal κ^* at three regulator weights is:

$\omega_{PD} = 0.40$ (execution-weighted): $\kappa^* = 3.0$ (corner)

$\omega_{PD} = 0.55$ (neutral baseline): $\kappa^* = 3.0$ (corner)

$\omega_{PD} = 0.70$ (discovery-weighted): $\kappa^* \approx 2.95$ (essentially corner)

$\omega_{PD} = 0.85$: $\kappa^* \approx 2.20$ (interior)

$\omega_{PD} = 0.95$: $\kappa^* \approx 1.44$ (interior)

The threshold ω_{PD}^* at which interior solutions first appear, on Bybit's calibration, is approximately 0.69. This is a much higher threshold than under the paper's canonical calibration ($\omega_{PD}^* \approx 0.42$, where the venue's microstructure had a more balanced $D_v / \sigma_q = 0.71$). On Bybit, the welfare-maximising design at any neutral or execution-leaning regulator weight is at the operational ceiling $\kappa_{\max}$, not at an interior welfare optimum.

C.5 Implications

First, the qualitative results of the paper are unchanged. Proposition 0's structural ceiling $\bar{\kappa}$ still bounds any feasible κ . Proposition 5's existence-of-interior-optimum still holds for sufficiently high ω_{PD} (now ≥ 0.69 on Bybit, was ≥ 0.42 under canonical calibration). The state-dependent result of Proposition 6 still holds — under cascade stress, $\kappa^*(s)$ rises and locks onto the operational ceiling. The corner-solution outcome on Bybit at the neutral baseline is, in fact, the same kind of corner solution that Proposition 6 predicts under cascade stress, except that for Bybit's microstructure the corner regime applies at all stress levels.

Second, the headline number $\kappa^* \approx 1.56$ of the paper's canonical calibration is venue-specific. It applies to a hypothetical venue with deeper visible books relative to flow (D_v / σ_q in the $[0.4, 0.7]$ interior band). Real perpetual venues — at least Bybit BTC-PERP — operate with much shallower books ($D_v / \sigma_q \approx 0.1$), and the corner solution dominates. A venue's calibrated κ^* is therefore strongly dependent on its own microstructure, and the paper's headline number should not be read as a universal prescription.

Third, the operational ceiling $\kappa_{\max}$ of §6 and §7.3.3 is the binding constraint in practice for Bybit-style venues. The paper's recommended defaults — calibration-anchored $\kappa^* = 1.0$, minimal-intervention $\kappa^* = 0.5$ — both lie far below Bybit's calibrated corner value ($\kappa^* = 3.0$ at $\omega_{PD} = 0.55$), and far below the structural ceiling $\bar{\kappa}$. This means the operational caps in current practice are themselves conservative relative to the model's prescription. There is room for venues to relax their hidden-depth caps from current levels toward the operational ceiling, and welfare gains accumulate across that range.

Fourth, the empirical D_v / σ_q ratio is itself a useful regime indicator. Venues with $D_v / \sigma_q < 0.4$ are in the corner-solution regime at typical regulator weights and should size their operational ceiling $\kappa_{\max}$ as the binding parameter; venues with $D_v / \sigma_q \in [0.4, 0.7]$ admit interior optima at neutral regulator weights and can use the canonical $\kappa^* \approx 1.56$ as a starting point; venues with $D_v / \sigma_q > 0.7$ may have welfare optima below 1.0. We propose this ratio as a venue-classification statistic for future empirical work across multiple perpetual venues.

C.6 Data-cleaning and robustness

Data cleaning. Trade records are taken from Bybit's public WebSocket trade feed (V5 API) and stored without filtering aside from deduplication on (trade_id, trade_time). Block trades, cross-venue trades, and self-matches are not separately tagged in the public feed; they are therefore included in the size statistics. We do not exclude liquidation-tagged trades (these are not consistently labelled in the public tape; in private venue data they would be excluded for the non-liquidation calibration). Book snapshots are level-2 aggregations sampled at approximately 1-second intervals, with each snapshot storing the top 25 levels per side as JSONB.

Tail-fit robustness. The Pareto MLE $\xi = 1.937$ is computed with $x_{\min}$ set to the empirical $D_v = 3.49$ BTC. Sensitivity to the choice of $x_{\min}$: at $x_{\min} = 5$ BTC ($n = 162$ big trades), ξ rises to approximately 2.1; at $x_{\min} = 2$ BTC ($n = 605$), ξ drops to approximately 1.7. The Hill plot is moderately stable over thresholds in $[2, 8]$ BTC, indicating that the Pareto fit is reasonable for the bulk of the tail but somewhat sensitive at the extremes. A more rigorous calibration would use a generalised Pareto distribution above an automatically-selected threshold; we use the simpler Hill estimator for transparency.

Sensitivity of the corner-solution finding. The corner-solution conclusion at neutral baseline depends primarily on the ratio $D_v / \sigma_q \approx 0.094$, not on the precise value of ξ . Re-running the welfare optimisation with $\xi \in [1.5, 2.5]$ (covering the Pareto-fit uncertainty above and the canonical 2.5) gives a corner solution at $\omega_{PD} = 0.55$ in all cases. The interior-optimum threshold ω_{PD}^* shifts from 0.69 (at $\xi = 1.937$) to approximately 0.62 (at $\xi = 2.5$) and 0.74 (at $\xi = 1.5$), but the qualitative finding — that Bybit's microstructure puts the welfare optimum at the corner under any neutral-or-execution-leaning regulator weight — is robust to the tail-shape uncertainty.

Window length. 18 hours of data is short for a calibration intended to inform venue policy. Longer windows (1–3 months) would smooth over diurnal volume variation, weekend effects, and macro news cycles. The headline empirical statistics ($D_v \approx 3.5$ BTC, $\sigma_q \approx 37$ BTC/min, $D_v / \sigma_q \approx 0.1$, $\xi \approx 1.9$) are likely robust to longer windows for a venue with Bybit's general activity profile, but precise quantitative values may shift. The corner-solution conclusion at the neutral baseline survives mechanically as long as D_v / σ_q remains below approximately 0.4 (the canonical interior-optimum band lower bound), which

appears to hold structurally on Bybit and likely on most major perpetual venues with similarly shallow inside books.

Walk-price counterfactual. The $\hat{\kappa}$ estimator (separate to the welfare calibration above; see `kappa_hat_estimates` table) infers hidden depth from the gap between observed fill volume and the visible depth that would have been required to absorb a trade absent hidden absorption. This walk-price counterfactual depends on the assumption that visible depth at the moment of fill is observable from the snapshot immediately preceding the trade; in practice, snapshots may lag fills by tens of milliseconds, leading to small under-estimation bias. We do not correct for this bias in the descriptive $\hat{\kappa}$ statistics; for the welfare calibration above, we use only D_v (averaged over snapshots) and σ_q (computed from trade tape only), neither of which depends on the walk-price counterfactual.

Single-venue scope. Only Bybit BTC-PERP is calibrated. Other major perpetual venues (Binance, OKX, dYdX, Hyperliquid) likely have different D_v / σ_q ratios, particularly DeFi venues where on-chain matching constraints alter inside-depth provision. Cross-venue comparison is the natural next step.

Liquidation flow. The trades dataset does not separately tag liquidation flow vs. ordinary informed-and-noise flow. The Pareto fit $\xi = 1.94$ averages across both populations. Under the framework of §7.3.2, separating liquidation from non-liquidation flow would yield a regime-specific ξ pair (lower ξ in liquidation flow); we do not have the venue-internal data needed to do this separation here. Future work using direct exchange data feeds with liquidation tagging would refine this.

Identification of β , σ_v , σ_u individually. We identified σ_q , D_v , and ξ — the three primitives that directly determine the welfare components $I(\kappa)$ and $E(\kappa)$ up to a normalisation constant. Separately identifying the underlying β , σ_v , σ_u (Kyle's structural parameters) requires an additional moment, typically the realised price-impact slope λ , which we did not estimate here. The welfare optimisation is invariant to this further identification because the relevant ratio σ_q^2 appears as a single object in the closed-form expressions of Proposition 2.

End of Appendix C.